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Constructing Quasi-Site-Specific Multivariate Probability Distribution Using Hierarchical Bayesian Model

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Abstract: In geotechnical engineering, it is challenging to construct a site-specific multivariate probability distribution model for soil/rock properties because the site-specific data are usually sparse and incomplete. In contrast, there are abundant generic soil/rock data in the literature for the construction of a generic multivariate probability distribution model, but this model is typically biased and/or imprecise for a specific site. A hybridization method has been proposed to combine these two sources of soil/rock data (site-specific data and a generic database) to produce a quasi-site-specific model, but this method is essentially heuristic. In the current paper, a more rational method that exploits the geologic origin of soil/rock data is proposed. There is a tendency for data to be more similar within a single site and less similar between sites. This is called site uniqueness in geotechnical engineering practice, but no data-driven methods exist to quantify this data feature currently. The hierarchical Bayesian model (HBM) is a natural model to exploit this group information. The grouping criterion can be site localization, soil/rock types, or others. This paper only studies the group criterion based on site localization. This means that a generic database is now viewed as a collection of data groups labeled by qualitative site labels. This site label does not contain any quantitative information such as GPS location, it merely demarcates each group as distinct. The novel contribution is the development of an efficient HBM with closed-form conditional probabilities based on suitably chosen conjugate priors that can handle multivariate, uncertain and unique, sparse, incomplete, and potentially corrupted (MUSIC) data containing site labels. Numerical comparisons between the hybridization method (which cannot incorporate group information) and HBM show that even the simple qualitative knowledge that data belong to a geographically constrained site can improve the estimation of soil/rock properties. The GPS location of each site is not needed. **DOI: 10.1061/(ASCE)EM.1943-7889.0001964.** © *2021 American Society of Civil Engineers.*

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Introduction

There are two relatively unique features of geotechnical site investigation data that could be juxtaposed in the context of data-driven decision-making. On one hand, when it comes to a local site, site-specific data are sparse and incomplete. Generic transformation models founded on data from multiple sites are common for this reason (Kulhawy and Mayne 1990). Phoon et al. (2019) described this situation as MUSIC: site-specific data are Multivariate, Uncertain and Unique, Sparse, Incomplete, and potentially Corrupted. Geotechnical engineers routinely need to make decisions based on MUSIC site-specific data. On the other hand, it is widely known that generic (non-site-specific) data are abundant. Phoon et al. (2019) coined the phrase *big indirect data* (BID) to emphasize that the common perception of data sparsity in geotechnical engineering is only accurate within a site-specific context. It is quite extreme to ignore BID; the implied assumption that all non-site-specific data are irrelevant is not aligned to current practice. Generic transformation models are widely used (Kulhawy and Mayne 1990). The site challenge is how to complement current practice steeped in empiricism with data-driven methods to extract maximum value from BID. To rally more interest in data-driven research, Phoon (2020) named this project AlphaGeo, given that the impetus comes from the success of Google's AlphaGo. Phoon and Ching (2021) focused more narrowly on data-driven subsurface mapping, and they named the project DeepGeo (inspired by DeepMind that produces AlphaGo).

The conventional and by far the most ubiquitous approach for estimating a soil/rock property from a measurement (say undrained shear strength from cone tip resistance) is regression [e.g., Kulhawy and Mayne (1990), Mayne et al. (2001), and Zhang (2016) compiled a large collection of such regression equations]. The underlying model is a bivariate probability density function (PDF) model. It is possible to estimate a soil property using multiple measurements (say, undrained shear strength from cone tip resistance, overconsolidation ratio, and liquidity index). The supporting model generalizes naturally to a multivariate PDF model. The current practice is to characterize this PDF model using a generic database (BID). However, generic data are indirect, so the resulting generic model may not be unbiased or precise for a particular site (Ching 2018). Specifically, its average trend may not be the same as the local average trend. Higher-order differences are likely, such as the transformation uncertainty about this average (first-order) trend. The ideal practice is to characterize a local model using site-specific data *alone*, but given the typical data-poor site-specific scenario, the statistical uncertainty can be significant and cannot be neglected. This local approach is widely advocated in practice, but there is

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no means of carrying this out thus far in routine projects. To our knowledge, all building regulations worldwide mandate a fixed minimum amount of site investigation regardless of the amount of data available in adjacent sites. For example, building regulations typically mandate minimum ground investigation at a specific site based on a relative or an absolute minimum number of boreholes. The Building and Construction Authority of Singapore (BCA) specified that the number of boreholes should be the greater of (1) one borehole per 300 m^2; or (2) one borehole at every interval between 10 and 30 m, but no less than three boreholes in a project site (BCA/IES/ACES Advisory Note 1/03 on Site Investigation and Load Tests). There is no reference to the amount of information already gathered in past neighboring projects. This is a testament of the significant weight assigned to site data, because BID is seen at most as a first-order guide, given its uncertain applicability to the site of interest. Here, the term *uncertain* leans toward *unknown unknowns* rather than *known unknowns* that could be quantified probabilistically. This issue has to some extent impeded the adoption of reliability methods in geotechnical engineering. In this sense, the site challenge is one of the pivotal questions that could move the needle in digitalizing existing practice if a satisfactory solution could be found.

Recently, Ching and Phoon (2019) proposed a heuristic method of constructing a quasi-site-specific multivariate PDF model by hybridizing a site-specific model (based on MUSIC data) with a generic model (based on BID). Ching and Phoon (2020) further incorporated the spatial correlation (MUSIC-X), so the site-specific model not only characterizes the cross-correlation among soil properties but also characterizes the spatial correlation within the target site. The idea of hybridization is straightforward. Fig. 1 illustrates the idea by the correlation relationship between $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v, where q_t = (corrected) cone tip resistance of cone penetration test (CPT), σ_v = vertical total stress; σ'_v = vertical effective stress; and s_u = undrained shear strength. This hybridization is based on the assumption that there exists a generic population, and the generic data from this population [smaller dots in Fig. 1(a)] are used to construct the generic PDF [dashed ellipse in Fig. 1(b)]. There is also a target-site population, and the target-site data [larger circles in Fig. 1(a)] are used to construct the target-site PDF [middle-sized ellipse in Fig. 1(b)]. The hybrid (quasi-site-specific) PDF is taken to be proportional to the direct product between the generic PDF and target-site PDF [smallest ellipse in Fig. 1(b)]. The hybrid PDF has the following desirable behavior (Ching and Phoon 2019):

1. When the target-site data are very sparse [e.g., Fig. 2(a)], the target-site PDF is very flat and close to a constant function. As a result, (hybrid PDF) is proportional to (generic PDF) × (target-site PDF) ≈ (generic PDF) × (constant function). Hence, (hybrid PDF) ≈ (generic PDF). This behavior of the hybrid PDF is reasonable: if local experience is absent, a reasonable choice is to rely on generic experience.
2. When the target-site data are abundant [e.g., Fig. 2(b)], the target-site PDF is relatively peaked. Note that the generic PDF is usually relatively flat compared with the target-site PDF because the generic PDF has to accommodate soil/rock data with a wide variety of geologic origins. Hence, the hybrid PDF tends to be close to the target-site PDF. This behavior of the hybrid PDF is also reasonable: if local experience is abundant, a reasonable choice is to rely on local experience.

This argument supporting the hybridization method is purely heuristic. However, this simple hybridization method invoking almost no assumption on the structure of the generic database has been shown to be reasonable in practice (Ching and Phoon 2019; Ching et al. 2020a, b), and it should be viewed as a baseline approach that other approaches can be compared against. Other approaches will invoke some structural assumptions, such as the concept of a group described later.

There is a geologic origin to soil/rock data. Geotechnical engineers have argued that each site is unique by appealing to this origin. However, there is no complete understanding in quantitative terms of how unique or which aspects are unique with reference to the MUSIC database. Nonetheless, partial empirical evidence exists. Fig. 3(a) replots the data in Fig. 1(a) with site labels appended. Data points from the same site are plotted with the same site label. The size of a site is geographical constrained by the typical size of a single building or building complex construction project, occupying a few football fields. Different tests are conducted within the boundaries of a site as defined previously. The results obtained from these tests conducted at different depths

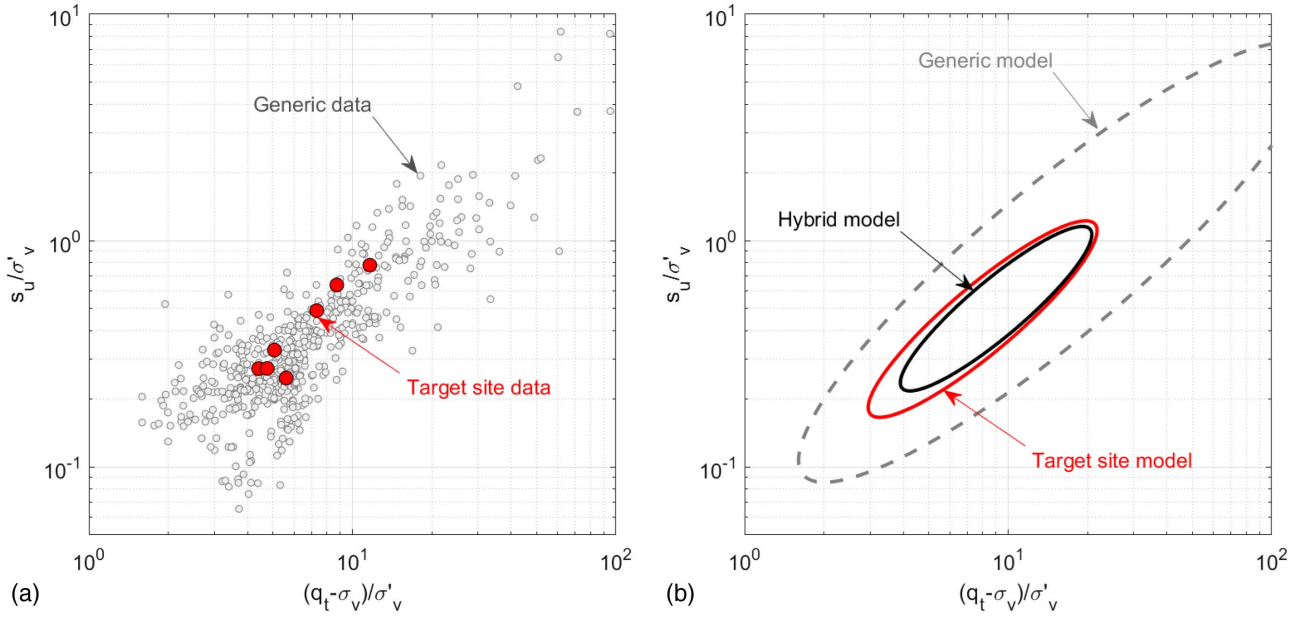


Fig. 1. (a) Generic versus target-site population; and (b) illustration for the hybridization idea.

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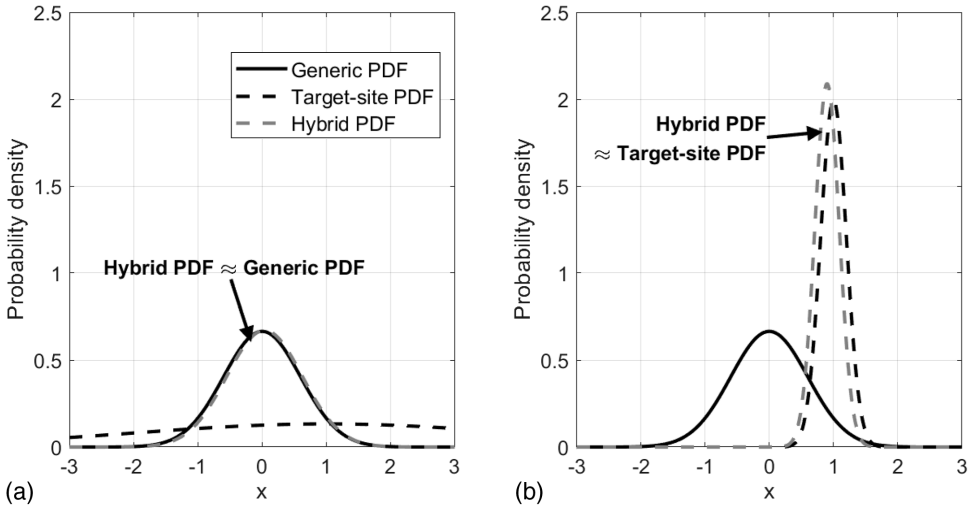


Fig. 2. Illustration for hybridization: (a) sparse target-site data; and (b) abundant target-site data. (Adapted from Ching and Phoon 2019.)

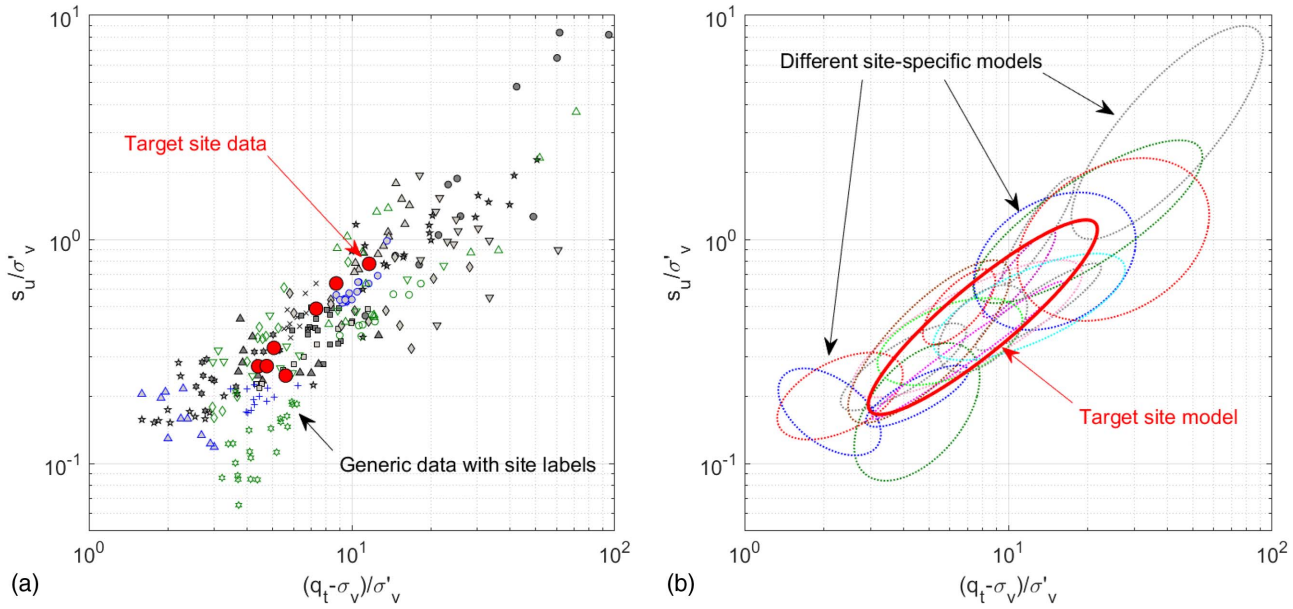


Fig. 3. (a) Replot of Fig. 1(a) with different sites differentiated by different markers (site labels); and (b) illustration for different site-specific models.

and different locations are grouped into a site. There are many sites ($i = 1, 2, \ldots, M$) in the generic database. Besides using site labels to group data, there can be other grouping criteria, e.g., rocks are customarily classified as igneous, metamorphic, and sedimentary (namely, rock-type groups). In the current paper, only the site label is used to group the data. This criterion is termed *site localization*, referring to data localized geographically within a project site as a group. Each group can be labeled by a unique site label (or marker) as shown in Fig. 3(a). The geographic size of the site (or the largest distance between two measurements) and its actual or even approximate location (e.g., nearest city) are not considered. The term *same site* or *different sites* is used in this broad qualitative sense from here on. Other reasonable grouping criteria such as location (GPS, city, region, country, etc.), geologic formation, and soil/rock type can be adopted, although not considered in the current study. One may contemplate optimizing the grouping criterion or even allowing it to emerge from the data by unsupervised learning techniques such as clustering, which is a challenging problem that deserves separate future research by itself and is not the focus of the current study. If the multivariate normal PDF model is assumed (to be elaborated on later), data points from the ith site in the database are assumed to be governed by their site-specific mean vector $\underline{\mu}_i$ and site-specific covariance matrix $\mathbf{C}_i$. There are many ($i = 1, 2, \ldots, M$) group- or site-specific models, illustrated as the dashed ellipses in Fig. 3(b). Data points from the target site are governed by their site-specific $\underline{\mu}_s$ and $\mathbf{C}_s$, illustrated as the solid ellipse in Fig. 3(b). There is nothing special about a target site. It is just one of the sites, but the subscript s is adopted to highlight its special status as the site of interest. One can imagine that this is the site where a new project is to be initiated and soil/rock parameters must be estimated to design the geotechnical structures. The data structure in Fig. 3 is featured by clusters of distributions. This structure is referred to as the *hierarchical structure* from here on. It is expected that the additional site-label information can produce extra value to a model if the model has the hierarchical structure. The hybridization method does not have the capacity to extract this information, because it does not have the hierarchical structure.

The concept regarding a generic database as being composed of multiple groups is referred to as the heterogeneity of data in the statistics community. A standard way to handle this situation is to

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use a hierarchical Bayesian model (HBM) to capture the underlying structure within the heterogeneous data (Gelman and Hill 2006). Interest in HBM is increasing in civil engineering (e.g., Behmanesh et al. 2015; Nagel and Sudret 2016; Yu and Baroud 2019) or even in geotechnical engineering (e.g., Zhang et al. 2014, 2016; Lu et al. 2018; Bozorgzadeh et al. 2019; Bozorgzadeh and Bathurst 2019; Ching 2020). While conceptually elegant, Wu et al. (2019) illustrated that HBM requires significant computational power to deal with complex probabilistic models without analytical Bayesian solutions. This may hinder the use of HBM for practical problems where complex models are needed. The issue of significant computational demand of HBM is likely to become more and more challenging as BID is expected to grow quickly in tandem with rapidly advancing digital technologies. It is important to emphasize that an actual generic geotechnical database does not need to follow HBM (just as spatial data do not need to follow a random field model), but HBM is a natural starting point to address this so-called "site recognition" challenge (Phoon et al. 2021), given the lack of past research on site uniqueness from a data-driven perspective.

The current paper proposes a novel HBM to deal with the soil/rock data in the presence of site labels in Fig. 3 to construct a quasi-site-specific model. With the novel HBM, the data from different sites are assumed to follow the hierarchical structure (i.e., clusters of distributions), and each site is characterized by its site-specific parameters $(\underline{\mu}_i, \mathbf{C}_i)$. These site-specific model parameters are assumed to be "similar but not identical" (Bozorgzadeh et al. 2019). Specifically, $\underline{\mu}_i \neq \underline{\mu}_j$ if $i \neq j$, but $\underline{\mu}_i$ and $\underline{\mu}_j$ are governed by the same hyperparameters. The hyperparameters govern the intersite mean of $\underline{\mu}_i$ as well as the intersite variability in $\underline{\mu}_i$. There are other hyperparameters that govern the intersite mean of $\mathbf{C}_i$ as well as the intersite variability in $\mathbf{C}_i$.

To make HBM practically useful, it is essential to mitigate the significant computational demand associated with complex probabilistic models without analytical Bayesian solutions. To resolve this computational-demand issue, the current paper assumes multivariate normality together with conjugate priors such that analytical Bayesian solutions exist for certain conditional probabilities. To our knowledge, this contribution is novel and critical to dealing with large actual geotechnical databases containing site-label information. Because analytical Bayesian solutions exist for the conditional probabilities, the Bayesian problem can be efficiently solved by adopting the Gibbs sampler. This paper may tilt the balance of HBM in serving as a theoretical tool for ideal problems to a practical tool that can deal with large actual geotechnical databases. The purpose of the current paper is threefold:

1. Propose the novel HBM with closed-form conditional probabilities and present its model structure.
2. Develop efficient learning and inference algorithms for the HBM.
3. Demonstrate the effectiveness of the HBM for constructing the quasi-site-specific model and compare it with the baseline hybridization method. Based on the comparison results, the value of the site-label information is demonstrated. *Site-label information* refers to the qualitative knowledge that the data are measured within a typical project site, i.e., geographical/site localization of data within, say, a building footprint. It does not refer to the physical site locations, exact or approximate.

Generic Soil Property Database

In the current paper, a generic clay property database named CLAY/10/7490 (Ching and Phoon 2014b) is adopted for demonstration. The CLAY/10/7490 database consists of 7,490 data points for 10 clay parameters from 251 studies in the literature that cover 30 countries/regions worldwide. To simplify the demonstration, only 2 out of the 10 clay parameters are considered for the time being: $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v. Fig. 4(a) shows the generic correlation relationship between $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v in the database, which shows a distinct generic trend with positive correlation. The larger dots in the figure are the data points for a particular site in CLAY/10/7490. It is clear that the site-specific statistics (for the larger dots) can be quite different from the generic statistics (for the smaller dots).

In the current study, the logarithms for $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v are denoted by y_1 and y_2, respectively

$$y_1 = \ln[(q_t - \sigma_v)/\sigma'_v]$$
$$y_2 = \ln(s_u/\sigma'_v) \tag{1}$$

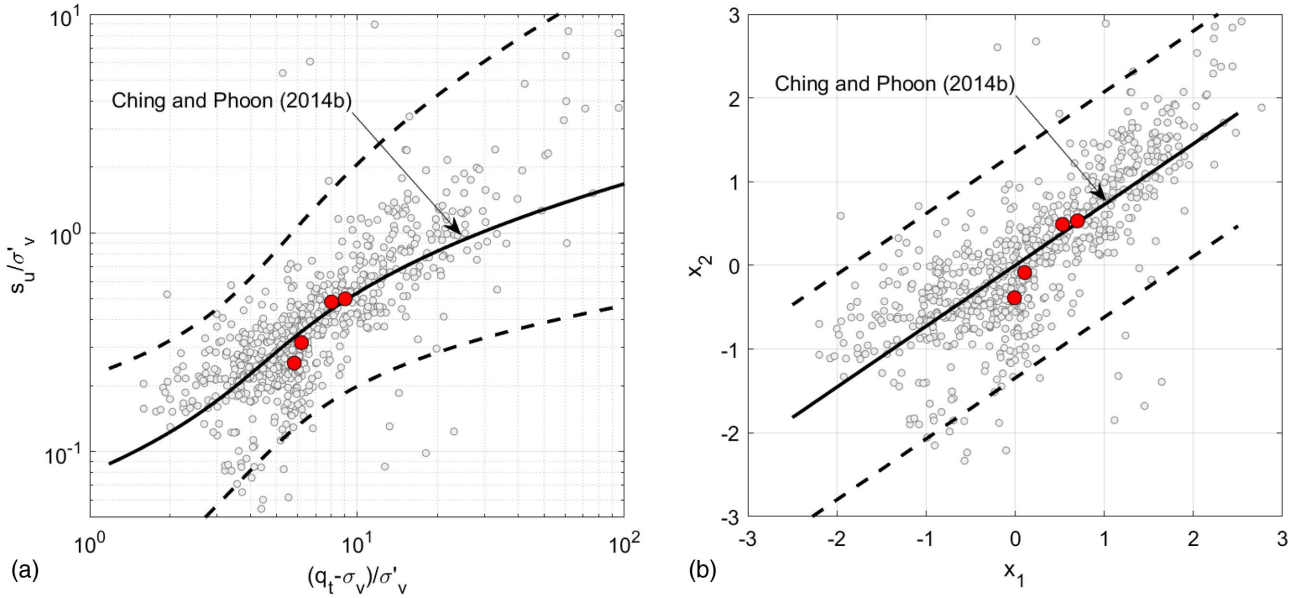


Fig. 4. (a) Correlation between $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v; and (b) correlation between x_1 and x_2.

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Ching and Phoon (2014a) fit the y_1 and y_2 data in CLAY/10/7490 by the Johnson SU distribution (Johnson 1949). The Johnson SU variables (y_1, y_2) can be converted into standard normal variables (x_1, x_2) by the following equations (Ching and Phoon 2014a):

$$\frac{x_1 + 0.572}{1.340} = \sinh^{-1}\left(\frac{y_1 - 1.476}{0.659}\right)$$
$$\frac{x_2 + 0.517}{2.039} = \sinh^{-1}\left(\frac{y_2 + 1.461}{1.427}\right) \quad (2)$$

Fig. 4(b) shows the correlation relationship between x_1 and x_2. Ching and Phoon (2014a) further assumed (x_1, x_2) to be bivariate normal. Under this assumption, the conditional median and 95% confidence interval (CI) for x_2 given the information of x_1 have analytical expressions. Fig. 4(b) shows the resulting conditional median and 95% CI for x_2 given the information of x_1, and they are mapped to the y-space in Fig. 4(a).

HBM with Closed-Form Conditional Probabilites

The current paper proposes a novel HBM to model the intersite mean and intersite variability in the generic database *in closed-form conditional probabilities*. The graphical illustration of the proposed HBM is shown in Fig. 5. The proposed HBM assumes that the ith site in the generic database follows its own site-specific model characterized by $(\underline{\mu}_i, \mathbf{C}_i)$ and that different sites have different mean vectors and covariance matrices, i.e., $(\underline{\mu}_i, \mathbf{C}_i) \neq (\underline{\mu}_j, \mathbf{C}_j)$ if $i \neq j$. Namely, the proposed HBM assumes that the generic database follows the hierarchical structure, i.e., clusters of distributions. Nonetheless, all site-specific mean vectors and covariance matrices are governed by the same underlying distribution parameterized by a set of hyperparameters.

Hierarchical Bayesian Modeling

Classical Bayesian analysis with a nonhierarchical model begins with assigning a model to the likelihood $f(\mathbf{X}|\underline{\mu}, \mathbf{C})$, which evaluates the probability density to observe the data $\mathbf{X}$ given the generic model parameters $(\underline{\mu}, \mathbf{C})$, and the prior $f(\underline{\mu}, \mathbf{C})$ represents our initial belief of the possible values of $(\underline{\mu}, \mathbf{C})$. Using Bayes' theorem, the posterior PDF of $(\underline{\mu}, \mathbf{C})$ can be expressed as

$$f(\underline{\mu}, \mathbf{C}|\mathbf{X}) = \frac{f(\mathbf{X}|\underline{\mu}, \mathbf{C}) f(\underline{\mu}, \mathbf{C})}{f(\mathbf{X})} \quad (3)$$

In contrast to classical Bayesian analysis, the hierarchical Bayesian model in Fig. 5 allows a more complicated modeling of $\mathbf{X}$. Specifically, the data $\mathbf{X}$ are formed by clusters of distributions, and the data for the ith cluster (site) are denoted by $\mathbf{X}_i$. Each site has its own model parameters $(\underline{\mu}_i, \mathbf{C}_i)$, and the prior $f(\underline{\mu}_i, \mathbf{C}_i|\mathbf{\Theta})$ is parametrized by the hyperparameters $\mathbf{\Theta}$. Suppose that there are M sites. The priors $f(\underline{\mu}_1, \mathbf{C}_1|\mathbf{\Theta}), f(\underline{\mu}_2, \mathbf{C}_2|\mathbf{\Theta}), \ldots$, and $f(\underline{\mu}_M, \mathbf{C}_M|\mathbf{\Theta})$ are parametrized by the same hyperparameters $\mathbf{\Theta}$. Using Bayes' theorem, the posterior PDF of $(\underline{\mu}_1, \ldots, \underline{\mu}_M, \mathbf{C}_1, \ldots, \mathbf{C}_M, \mathbf{\Theta})$ can be expressed as

$$f(\underline{\mu}_1, \ldots, \underline{\mu}_M, \mathbf{C}_1, \ldots, \mathbf{C}_M, \mathbf{\Theta}|\mathbf{X}) = \frac{[\prod_{i=1}^{M} f(\mathbf{X}_i|\underline{\mu}_i, \mathbf{C}_i) f(\underline{\mu}_i, \mathbf{C}_i|\mathbf{\Theta})] f(\Theta)}{f(\mathbf{X})} \quad (4)$$

In our study, we focus on the posterior PDF $f(\mathbf{\Theta}|\mathbf{X})$

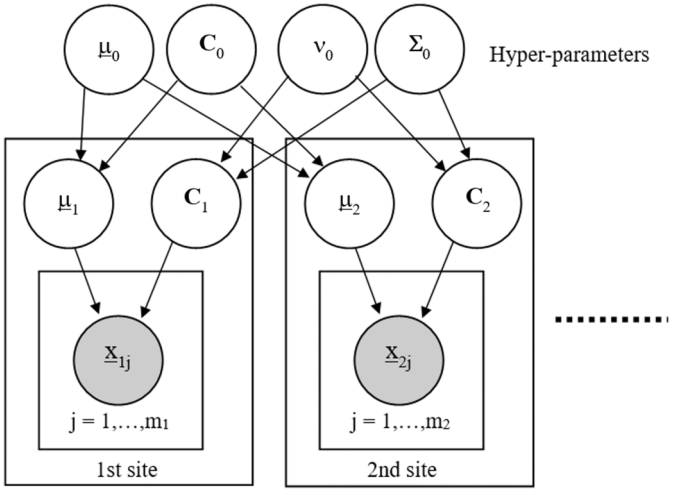


Fig. 5. Graphical illustration of the proposed HBM.

$$f(\mathbf{\Theta}|\mathbf{X}) = \frac{[\prod_{i=1}^{M} \int f(\mathbf{X}_i|\underline{\mu}_i, \mathbf{C}_i) f(\underline{\mu}_i, \mathbf{C}_i|\mathbf{\Theta}) d\underline{\mu}_i d\mathbf{C}_i] f(\mathbf{\Theta})}{f(\mathbf{X})} \quad (5)$$

Because the Bayesian analysis for $f(\mathbf{\Theta}|\mathbf{X})$ involves a potentially high dimensional integral, it is important to use models such that an analytical solution exists for the integral.

Model Structure

To be more specific, the proposed HBM assumes the following generative process for the data points of the ith site in the database:

1. Generate the site-specific mean vector $\underline{\mu}_i \sim f(\underline{\mu}_i|\underline{\mu}_0, \mathbf{C}_0) = \mathrm{N}(\underline{\mu}_i; \underline{\mu}_0, \mathbf{C}_0)$, a multivariate normal PDF with mean vector = $\underline{\mu}_0$ and covariance matrix = $\mathbf{C}_0$. $(\underline{\mu}_0, \mathbf{C}_0)$ are the hyperparameters that govern the distribution of $\underline{\mu}_i$ across all the sites in the generic database: $\underline{\mu}_0$ governs the mean of $\underline{\mu}_i$, whereas $\mathbf{C}_0$ governs the variability in $\underline{\mu}_i$.
2. Generate the site-specific covariance matrix $\mathbf{C}_i \sim f(\mathbf{C}_i|\mathbf{\Sigma}_0, \nu_0) = \mathrm{IW}(\mathbf{C}_i; \mathbf{\Sigma}_0, \nu_0)$, an inverse-Wishart PDF (James 1964; Mardia et al. 1979) with scale matrix=$\mathbf{\Sigma}_0$ and degree of freedom = $\nu_0 \cdot (\mathbf{\Sigma}_0, \nu_0)$ are the hyperparameters that govern the distribution of $\mathbf{C}_i$ across all the sites in the generic database.
3. Let $\underline{x}_{ij}$ denote the jth data point in the ith site in the database ($j = 1, \ldots, m_i$; m_i is the number of data points for the ith site). Generate $\underline{x}_{ij} \sim \mathrm{N}(\underline{x}_{ij}; \underline{\mu}_i, \mathbf{C}_i)$.

The hyperparameters $\mathbf{\Theta} = (\underline{\mu}_0, \mathbf{C}_0, \nu_0, \mathbf{\Sigma}_0)$ play a key role in the proposed HBM. They govern how site-specific model parameters $\{(\underline{\mu}_i, \mathbf{C}_i): i = 1, 2, \ldots, M\}$ are generated. Namely, they govern the intersite mean and intersite variability in $(\underline{\mu}_i, \mathbf{C}_i)$. The main purpose for the learning stage for the HBM (to be discussed later) is to learn $\mathbf{\Theta}$ based on a generic database containing site-label information. There is no reason to believe that site uniqueness or intersite variability can be fully captured by $\mathbf{\Theta}$, but this HBM offers a natural scaffolding to take a first stab at studying site uniqueness systematically in quantitative terms beyond sole reliance on engineering judgment. Note that the baseline hybridization method does not have the hierarchical structure, so it cannot learn the site-label information.

To complete the probabilistic model for the HBM, the following PDFs are required:

1. (Conditional PDF of $\underline{\mu}_i$) $\underline{\mu}_i$ is assumed to follow the following multivariate normal PDF:

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$$f(\underline{\mu}_i|\underline{\mu}_0, \mathbf{C}_0) = \mathrm{N}(\underline{\mu}_i; \underline{\mu}_0, \mathbf{C}_0) = |\mathbf{C}_0|^{-\frac{1}{2}} \cdot (2\pi)^{-\frac{n}{2}} \cdot \exp\left[-\frac{1}{2}(\underline{\mu}_i - \underline{\mu}_0)^T \mathbf{C}_0^{-1}(\underline{\mu}_i - \underline{\mu}_0)\right] \quad (6)$$

where n = variable dimension ($n = 2$ because there are two variables of interest, x_1 and x_2); and superscript T = matrix transpose.

- (Prior PDF of $\underline{\mu}_0$) In Bayesian analysis, it is necessary to specify the prior PDFs for the hyperparameters $\boldsymbol{\Theta} = (\underline{\mu}_0, \mathbf{C}_0, \nu_0, \boldsymbol{\Sigma}_0)$. For a given likelihood function $f(\text{data}|\theta)$, if the posterior PDF $f(\theta|\text{data})$ is in the same distribution family as the prior PDF $f(\theta)$, the prior PDF called a *conjugate prior* for the likelihood $f(\text{data}|\theta)$. To manage the computational demand for the proposed HBM, it is desirable to adopt conjugate priors for the hyperparameters $\boldsymbol{\Theta} = (\underline{\mu}_0, \mathbf{C}_0, \nu_0, \boldsymbol{\Sigma}_0)$ such that their posterior PDFs have closed forms. Table 1 gives the likelihood function associated with each hyperparameter as well as the corresponding conjugate prior and posterior. According to Table 1, the conjugate prior for $\underline{\mu}_0$ is a multivariate normal PDF, $\mathrm{N}(\underline{\mu}_0; \underline{\mu}_{\underline{\mu}_0}, \mathbf{C}_{\underline{\mu}_0})$. It is also desirable to adopt noninformative priors in the absence of prior information. This can be done by taking $\underline{\mu}_{\underline{\mu}_0}$ to be a zero vector and $\mathbf{C}_{\underline{\mu}_0}$ to be a diagonal matrix with larger diagonals (in this study, 10^4 is used).
- (Prior PDF of $\mathbf{C}_0$) According to Table 1, the conjugate prior for $\mathbf{C}_0$ is an inverse-Wishart (IW) PDF

$$f(\mathbf{C}_0|\boldsymbol{\Sigma}_{\mathbf{C}_0}, \nu_{\mathbf{C}_0}) = \mathrm{IW}(\mathbf{C}_0; \boldsymbol{\Sigma}_{\mathbf{C}_0}, \nu_{\mathbf{C}_0}) = \frac{|\boldsymbol{\Sigma}_{\mathbf{C}_0}|^{\nu_{\mathbf{C}_0}/2}}{2^{n\nu_{\mathbf{C}_0}/2} \cdot \Gamma_n(\nu_{\mathbf{C}_0}/2)} \cdot |\mathbf{C}_0|^{-\frac{\nu_{\mathbf{C}_0}+n+1}{2}} \cdot \exp\left[-\frac{1}{2} tr(\boldsymbol{\Sigma}_{\mathbf{C}_0} \times \mathbf{C}_0^{-1})\right] \quad (7)$$

where $\Gamma_n(\cdot)$ = multivariate Gamma function (James 1964) with dimension = n; and $tr(\cdot)$ = matrix trace, i.e., the summation of the diagonal elements of a square matrix. It is not straightforward to make an inverse-Wishart PDF noninformative. To achieve an approximately noninformative IW PDF, Huang and Wand (2013) proposed that $\nu_{\mathbf{C}_0} = n + 1$ and $\boldsymbol{\Sigma}_{\mathbf{C}_0} = 4 \times \mathrm{diag}(1/a_1, \ldots, 1/a_n)$ [diag$(\cdot)$ means a diagonal matrix], and a_k is random and distributed as an inverse-gamma (IG) PDF with shape parameter = α and scale parameter = β

$$f(a_k) = \mathrm{IG}(a_k; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot a_k^{-\alpha-1} \cdot \exp\left(-\frac{\beta}{a_k}\right) \quad (8)$$

Huang and Wand (2013) showed that if $\alpha = 0.5$ and β = a small number (in this study, 10^{-4} is adopted), $\mathrm{IW}(\mathbf{C}_0; \boldsymbol{\Sigma}_{\mathbf{C}_0}, \nu_{\mathbf{C}_0})$ is fairly noninformative.

2. (Conditional PDF of $\mathbf{C}_i$) $\mathbf{C}_i$ is assumed to follow the following inverse-Wishart PDF:

$$f(\mathbf{C}_i|\boldsymbol{\Sigma}_0, \nu_0) = \mathrm{IW}(\mathbf{C}_i; \boldsymbol{\Sigma}_0, \nu_0) = \frac{|\boldsymbol{\Sigma}_0|^{\nu_0/2}}{2^{n\nu_0/2} \cdot \Gamma_n(\nu_0/2)} \cdot |\mathbf{C}_i|^{-\frac{\nu_0+n+1}{2}} \cdot \exp\left[-\frac{1}{2} tr(\boldsymbol{\Sigma}_0 \times \mathbf{C}_i^{-1})\right] \quad (9)$$

where $(\boldsymbol{\Sigma}_0, \nu_0)$ = hyperparameters.

- (Prior PDF of $\boldsymbol{\Sigma}_0$) According to Table 1, the conjugate prior for $\boldsymbol{\Sigma}_0$ is a Wishart PDF

$$f(\boldsymbol{\Sigma}_0|\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}, \lambda_{\boldsymbol{\Sigma}_0}) = \mathrm{W}(\boldsymbol{\Sigma}_0; \boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}, \lambda_{\boldsymbol{\Sigma}_0}) = \frac{|\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}|^{-\lambda_{\boldsymbol{\Sigma}_0}/2}}{2^{n\lambda_{\boldsymbol{\Sigma}_0}/2} \cdot \Gamma_n(\lambda_{\boldsymbol{\Sigma}_0}/2)} \cdot |\boldsymbol{\Sigma}_0|^{\frac{\lambda_{\boldsymbol{\Sigma}_0}-n-1}{2}} \cdot \exp\left[-\frac{1}{2} tr(\boldsymbol{\Sigma}_0 \times \boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1})\right] \quad (10)$$

where $\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}$ = scale matrix; and $\lambda_{\boldsymbol{\Sigma}_0}$ = degree of freedom. Chung et al. (2015) showed that if $\lambda_{\boldsymbol{\Sigma}_0} = n + 2$ and $\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}$ is a diagonal matrix with large diagonals (in this study, 10^4 is adopted), $\mathrm{W}(\boldsymbol{\Sigma}_0; \boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}, \lambda_{\boldsymbol{\Sigma}_0})$ is fairly noninformative.

- (Prior distribution ν_0) ν_0 is a discrete variable ranging from n to infinity. To make the prior distribution of ν_0 noninformative, a uniform distribution between n and a large upper bound (in this study, 1,000 is adopted as the upper bound) is adopted.

3. (Conditional PDF of $\underline{x}_{ij}$) $\underline{x}_{ij}$ is assumed to follow the following multivariate normal PDF:

$$f(\underline{x}_{ij}|\underline{\mu}_i, \mathbf{C}_i) = \mathrm{N}(\underline{x}_{ij}; \underline{\mu}_i, \mathbf{C}_i) = |\mathbf{C}_i|^{-\frac{1}{2}} \cdot (2\pi)^{-\frac{n}{2}} \cdot \exp\left[-\frac{1}{2}(\underline{x}_{ij} - \underline{\mu}_i)^T \mathbf{C}_i^{-1}(\underline{x}_{ij} - \underline{\mu}_i)\right] \quad (11)$$

For $n = 2$ (the current example), $\underline{x}_{ij} = (\underline{x}_{1,ij}\, \underline{x}_{2,ij})^T$.

Let $\mathbf{X}$ contain all data points in the database, i.e., $\mathbf{X} = \{\underline{x}_{ij}: i = 1, \ldots, M; j = 1, \ldots, m_i\}$, where M is the total number of sites in the database, and m_i is the total number of soil/rock records at the ith site. Similarly, let $\boldsymbol{\mu} = \{\underline{\mu}_i: i = 1, \ldots, M\}$ and $\mathbf{C} = \{\mathbf{C}_i: i = 1, \ldots, M\}$. For the time being, we assume that all entries in $\mathbf{X}$ are known (no missing values). Later, the scenario with missing values in $\mathbf{X}$ will be addressed. We further assume x_{ij} and x_{ik} to be independent. In other words, we do not consider spatial correlation in MUSIC-X in this paper (Ching and Phoon 2019). A new method called HBM-MUSIC-3X has been proposed recently to address this gap (Ching et al. 2021, Phoon and Ching 2021). The variables in the proposed HBM include the transformed soil/rock data $\mathbf{X}$, site-specific normal model parameters $(\boldsymbol{\mu}, \mathbf{C})$, hyperparameters $\boldsymbol{\Theta}$, and $(a_1, a_2, \ldots, a_n)$. The complete multivariate PDF for all variables can be written as

Table 1. Conjugate priors and posteriors for the hyperparameters

Hyperparameter	Likelihood(s)	Conjugate distribution	Prior PDF	Posterior PDF
$\underline{\mu}_0$	$\mathrm{N}(\underline{\mu}_i; \underline{\mu}_0, \mathbf{C}_0)$ $(i = 1, \ldots, M)$	Normal	$\mathrm{N}(\underline{\mu}_0; \underline{\mu}_{\underline{\mu}_0}, \mathbf{C}_{\underline{\mu}_0})$	$N\{\underline{\mu}_0; (\mathbf{C}_{\underline{\mu}_0}^{-1} + M \cdot \mathbf{C}_0^{-1})^{-1}(\mathbf{C}_0^{-1} \sum_{i=1}^M \underline{\mu}_i), (\mathbf{C}_{\underline{\mu}_0}^{-1} + M \cdot \mathbf{C}_0^{-1})^{-1}\}$
$\mathbf{C}_0$	$\mathrm{N}(\underline{\mu}_i; \underline{\mu}_0, \mathbf{C}_0)$ $(i = 1, \ldots, M)$	Inverse-Wishart	$\mathrm{IW}(\mathbf{C}_0; \boldsymbol{\Sigma}_{\mathbf{C}_0}, \nu_{\mathbf{C}_0})$	$\mathrm{IW}\{\mathbf{C}_0; \boldsymbol{\Sigma}_{\mathbf{C}_0} + \sum_{i=1}^M (\underline{\mu}_i - \underline{\mu}_0)(\underline{\mu}_i - \underline{\mu}_0)^T, M + \nu_{\mathbf{C}_0}\}$
$\boldsymbol{\Sigma}_0$	$\mathrm{IW}(\mathbf{C}_i; \boldsymbol{\Sigma}_0, \nu_0)$ $(i = 1, \ldots, M)$	Wishart	$\mathrm{W}(\boldsymbol{\Sigma}_0; \boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}, \lambda_{\boldsymbol{\Sigma}_0})$	$\mathrm{W}\{\boldsymbol{\Sigma}_0; \boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1} + \sum_{i=1}^M \mathbf{C}_i^{-1}, M\nu_0 + \lambda_{\boldsymbol{\Sigma}_0}\}$

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$$
\begin{aligned}
f(\mathbf{X},\boldsymbol{\mu},\mathbf{C},\Theta,\underline{a}) &= f(\mathbf{X}|\boldsymbol{\mu},\mathbf{C})\cdot f(\boldsymbol{\mu}|\underline{\mu}_0,\mathbf{C}_0)\cdot f(\mathbf{C}|\boldsymbol{\Sigma}_0,\nu_0)\cdot f(\underline{\mu}_0)\cdot f(\mathbf{C}_0)\cdot f(\boldsymbol{\Sigma}_0)\cdot p(\nu_0)\cdot f(\underline{a}) \\
&= \left[\prod_{i=1}^{M}\prod_{j=1}^{m_i}\mathrm{N}(\underline{x}_{ij};\underline{\mu}_i,\mathbf{C}_i)\right]\cdot\left[\prod_{i=1}^{M}\mathrm{N}(\underline{\mu}_i;\underline{\mu}_0,\mathbf{C}_0)\right]\cdot\left[\prod_{i=1}^{M}\mathrm{IW}(\mathbf{C}_i;\boldsymbol{\Sigma}_0,\nu_0)\right]\cdot\mathrm{N}(\underline{\mu}_0;\underline{\mu}_{\underline{\mu}_0},\mathbf{C}_{\underline{\mu}_0})\cdot\mathrm{IW}(\mathbf{C}_0;\boldsymbol{\Sigma}_{\mathbf{C}_0},\nu_{\mathbf{C}_0}) \\
&\quad\cdot\mathrm{W}(\boldsymbol{\Sigma}_0;\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0},\lambda_{\boldsymbol{\Sigma}_0})\cdot p(\nu_0)\cdot\left[\prod_{k=1}^{n}\mathrm{IG}(a_k|\alpha,\beta)\right] \\
&\propto \left(\prod_{i=1}^{M}|\mathbf{C}_i|^{-\frac{m_i+\nu_0+n+1}{2}}\right)\cdot|\mathbf{C}_0|^{-\frac{M+\nu_{\mathbf{C}_0}+n+1}{2}}\cdot|\boldsymbol{\Sigma}_0|^{\frac{M\nu_0+\lambda_{\boldsymbol{\Sigma}_0}-1}{2}}\cdot\frac{\prod_{k=1}^{n}a_k^{-\alpha-1}}{2^{Mn\nu_0/2}\cdot\Gamma_n(\nu_0/2)^M} \\
&\quad\cdot e^{-\frac{1}{2}\sum_{i=1}^{M}\sum_{j=1}^{m_i}(\underline{x}_{ij}-\underline{\mu}_i)^T\mathbf{C}_i^{-1}(\underline{x}_{ij}-\underline{\mu}_i)-\frac{1}{2}\sum_{i=1}^{M}(\underline{\mu}_i-\underline{\mu}_0)^T\mathbf{C}_0^{-1}(\underline{\mu}_i-\underline{\mu}_0)-\frac{1}{2}(\underline{\mu}_0-\underline{\mu}_{\underline{\mu}_0})^T\mathbf{C}_{\underline{\mu}_0}^{-1}(\underline{\mu}_0-\underline{\mu}_{\underline{\mu}_0})-\frac{1}{2}tr\left[\boldsymbol{\Sigma}_0\times\left(\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1}+\sum_{i=1}^{M}\mathbf{C}_i^{-1}\right)\right]-\frac{1}{2}tr(\boldsymbol{\Sigma}_{\mathbf{C}_0}\times\mathbf{C}_0^{-1})-\sum_{k=1}^{n}\frac{\beta}{a_k}}
\end{aligned}
\tag{12}
$$

where $p(\cdot)$ = discrete distribution; and $\underline{a}$ denotes $(a_1, a_2, \ldots, a_n)$.

Learning Stage: The Gibbs Sampler

The purpose of the learning stage is to learn the hyperparameters $\boldsymbol{\Theta} = (\underline{\mu}_0, \mathbf{C}_0, \nu_0, \boldsymbol{\Sigma}_0)$ based on the database $\mathbf{X}$. In the current paper, the Gibbs sampler (GS) is adopted for the learning. The GS (Geman and Geman 1984; Gilks et al. 1996) is a special instance for the Markov chain Monte Carlo methods (Gilks et al. 1996). The basic idea of the GS is to decompose the random parameters into groups and to randomly sample each group by conditioning on the remaining groups. By doing so sequentially among all groups, the samples are asymptotically distributed as the desired distribution. In our case, the random parameters include $(\mathbf{X},\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$. Although the closed-form for the full posterior PDF $f(\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a}|\mathbf{X})$ is not available, the following conditional PDFs have closed forms:

$$
\begin{aligned}
f(\underline{\mu}_i|\mathbf{X},\boldsymbol{\mu}_{\backslash i},\mathbf{C},\boldsymbol{\Theta},\underline{a}) &\propto e^{-\frac{1}{2}\sum_{j=1}^{m_i}(\underline{x}_{ij}-\underline{\mu}_i)^T\mathbf{C}_i^{-1}(\underline{x}_{ij}-\underline{\mu}_i)-\frac{1}{2}(\underline{\mu}_i-\underline{\mu}_0)^T\mathbf{C}_0^{-1}(\underline{\mu}_i-\underline{\mu}_0)} \\
&= \mathrm{N}\left\{\underline{\mu}_i;(\mathbf{C}_0^{-1}+m_i\mathbf{C}_i^{-1})^{-1}\left(\mathbf{C}_0^{-1}\underline{\mu}_0+\mathbf{C}_i^{-1}\sum_{j=1}^{m_i}\underline{x}_{ij}\right),(\mathbf{C}_0^{-1}+m_i\mathbf{C}_i^{-1})^{-1}\right\}
\end{aligned}
\tag{13}
$$

$$
f(\mathbf{C}_i|\mathbf{X},\boldsymbol{\mu},\mathbf{C}_{\backslash i},\boldsymbol{\Theta},\underline{a}) \propto |\mathbf{C}_i|^{-\frac{m_i+\nu_0+n+1}{2}}\cdot e^{-\frac{1}{2}\sum_{j=1}^{m_i}(\underline{x}_{ij}-\underline{\mu}_i)^T\mathbf{C}_i^{-1}(\underline{x}_{ij}-\underline{\mu}_i)-\frac{1}{2}tr(\boldsymbol{\Sigma}_0\times\mathbf{C}_i^{-1})} = \mathrm{IW}\left\{\mathbf{C}_i;\boldsymbol{\Sigma}_0+\sum_{j=1}^{m_i}(\underline{x}_{ij}-\underline{\mu}_i)(\underline{x}_{ij}-\underline{\mu}_i)^T,m_i+\nu_0\right\} \tag{14}
$$

$$
\begin{aligned}
f(\underline{\mu}_0|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\mathbf{C}_0,\nu_0,\boldsymbol{\Sigma}_0,\underline{a}) &\propto e^{-\frac{1}{2}\sum_{i=1}^{M}(\underline{\mu}_i-\underline{\mu}_0)^T\mathbf{C}_0^{-1}(\underline{\mu}_i-\underline{\mu}_0)-\frac{1}{2}(\underline{\mu}_0-\underline{\mu}_{\underline{\mu}_0})^T\mathbf{C}_{\underline{\mu}_0}^{-1}(\underline{\mu}_0-\underline{\mu}_{\underline{\mu}_0})} \\
&= \mathrm{N}\left\{\underline{\mu}_0;(\mathbf{C}_{\underline{\mu}_0}^{-1}+M\cdot\mathbf{C}_0^{-1})^{-1}\left(\mathbf{C}_{\underline{\mu}_0}^{-1}\underline{\mu}_{\underline{\mu}_0}+\mathbf{C}_0^{-1}\sum_{i=1}^{M}\underline{\mu}_i\right),(\mathbf{C}_{\underline{\mu}_0}^{-1}+M\cdot\mathbf{C}_0^{-1})^{-1}\right\}
\end{aligned}
\tag{15}
$$

$$
\begin{aligned}
f(\mathbf{C}_0|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\underline{\mu}_0,\nu_0,\boldsymbol{\Sigma}_0,\underline{a}) &\propto |\mathbf{C}_0|^{-\frac{M+\nu_{\mathbf{C}_0}+n+1}{2}}\cdot e^{-\frac{1}{2}\sum_{i=1}^{M}(\underline{\mu}_i-\underline{\mu}_0)^T\mathbf{C}_0^{-1}(\underline{\mu}_i-\underline{\mu}_0)-\frac{1}{2}tr(\boldsymbol{\Sigma}_{\mathbf{C}_0}\times\mathbf{C}_0^{-1})} \\
&= \mathrm{IW}\left\{\mathbf{C}_0;\boldsymbol{\Sigma}_{\mathbf{C}_0}+\sum_{i=1}^{M}(\underline{\mu}_i-\underline{\mu}_0)(\underline{\mu}_i-\underline{\mu}_0)^T,M+\nu_{\mathbf{C}_0}\right\}
\end{aligned}
\tag{16}
$$

$$
f(\boldsymbol{\Sigma}_0|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\underline{\mu}_0,\mathbf{C}_0,\nu_0,\underline{a}) \propto |\boldsymbol{\Sigma}_0|^{\frac{M\nu_0+\lambda_{\boldsymbol{\Sigma}_0}-1}{2}}\cdot e^{-\frac{1}{2}tr\left[\boldsymbol{\Sigma}_0\times\left(\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1}+\sum_{i=1}^{M}\mathbf{C}_i^{-1}\right)\right]} = \mathrm{W}\left\{\boldsymbol{\Sigma}_0;\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1}+\sum_{i=1}^{M}\mathbf{C}_i^{-1},M\nu_0+\lambda_{\boldsymbol{\Sigma}_0}\right\} \tag{17}
$$

$$
p(\nu_0|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\underline{\mu}_0,\mathbf{C}_0,\boldsymbol{\Sigma}_0,\underline{a}) \propto \frac{\prod_{i=1}^{M}|\mathbf{C}_i|^{-\frac{\nu_0}{2}}\cdot|\boldsymbol{\Sigma}_0|^{\frac{M\nu_0}{2}}}{2^{Mn\nu_0/2}\cdot\Gamma_n(\nu_0/2)^M} \tag{18}
$$

$$
f(a_k|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},_a_{\backslash k}) \propto a_k^{-(n+1)/2-\alpha-1}\cdot e^{-\frac{\beta+2\mathbf{C}_{0,kk}^{-1}}{a_k}} = \mathrm{IG}\left(a_k;\alpha+\frac{n+1}{2},\beta+2/\mathrm{C}_{0,kk}\right) \tag{19}
$$

where $\boldsymbol{\mu}_{\backslash i} = (\underline{\mu}_1, \ldots, \underline{\mu}_{i-1}, \underline{\mu}_{i+1}, \ldots, \underline{\mu}_M)$; $\mathbf{C}_{\backslash i} = (\mathbf{C}_1, \ldots, \mathbf{C}_{i-1}, \mathbf{C}_{i+1}, \ldots, \mathbf{C}_M)$; $\underline{a}_{\backslash k} = (a_1, \ldots, a_{k-1}, a_{k+1}, \ldots, a_n)$; and $\mathrm{C}_{0,kk}^{-1} =$ (k,k) entry in the $\mathbf{C}_0^{-1}$ matrix. These closed forms exist because of the assumption of multivariate normality with the conjugate priors. As a result, the GS can be conducted efficiently because *exact* sampling from one group conditional on the remaining groups is possible.

With the GS, it is also possible to accommodate missing values in $\mathbf{X}$. This incompleteness feature is prevalent in all geotechnical databases and must be considered in all data-driven methods. Let us

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denote the missing (unobserved) values in $\underline{x}_{ij}$ by $\underline{x}_{ij}^u$ and denote the observed values by $\underline{x}_{ij}^o$. The mean vector $\underline{\mu}_i$ and covariance matrix $\mathbf{C}_i$ are both partitioned accordingly: $\underline{\mu}_i^u$ is the submean vector in $\underline{\mu}_i$ for the unobserved components, and $\underline{\mu}_i^o$ is the submean vector in $\underline{\mu}_i$ for the observed components. Similarly, $\mathbf{C}_i^u$ is the subsquare matrix in $\mathbf{C}_i$ for the unobserved components, $\mathbf{C}_i^o$ is the subsquare matrix in $\mathbf{C}_i$ for the observed components, and $\mathbf{C}_i^{uo}$ is the subcovariance matrix in $\mathbf{C}_i$ for the covariance between $\underline{x}_{ij}^u$ and $\underline{x}_{ij}^o$. The PDF of $\underline{x}_{ij}^u$ conditioning on $\underline{x}_{ij}^o$ is still multivariate normal

$$f(\underline{x}_{ij}^u|\underline{x}_{ij}^o,\boldsymbol{\mu},\mathbf{C},\underline{\mu}_0,\mathbf{C}_0,\nu_0,\boldsymbol{\Sigma}_0,\underline{a}) = \mathrm{N}\{\underline{x}_{ij}^u;\underline{\mu}_i^u+\mathbf{C}_i^{uo}(\mathbf{C}_i^o)^{-1}(\underline{x}_{ij}^o-\underline{\mu}_i^o),\mathbf{C}_i^u-\mathbf{C}_i^{uo}(\mathbf{C}_i^o)^{-1}\mathbf{C}_i^{ou}\} \tag{20}$$

As a result, it is possible to simulate the missing values $\underline{x}_{ij}^u$ during the GS. There is no need to simulate $\underline{x}_{ij}^o$ because $\underline{x}_{ij}^o$ contains known data.

In summary, the GS iterates over the following steps:

1. Initialize $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$ samples at arbitrary values, where $\mathbf{X}^u$ denotes all missing values in $\mathbf{X}$. Note that $\mathbf{X}$ does not contain missing values after the initialization.
2. (For $i=1,\ldots,M$) Simulate $\underline{\mu}_i$ (MATLAB command "mvnrnd")

$$\underline{\mu}_i \sim \mathrm{N}\left\{\underline{\mu}_i;(\mathbf{C}_0^{-1}+m_i\mathbf{C}_i^{-1})^{-1}\left(\mathbf{C}_0^{-1}\underline{\mu}_0+\mathbf{C}_i^{-1}\sum_{j=1}^{m_i}\underline{x}_{ij}\right),(\mathbf{C}_0^{-1}+m_i\mathbf{C}_i^{-1})^{-1}\right\} \tag{21}$$

3. (For $i=1,\ldots,M$) Simulate $\mathbf{C}_i$ (MATLAB command "iwishrnd")

$$\mathbf{C}_i \sim \mathrm{IW}\left\{\mathbf{C}_i;\boldsymbol{\Sigma}_0+\sum_{j=1}^{m_i}(\underline{x}_{ij}-\underline{\mu}_i)(\underline{x}_{ij}-\underline{\mu}_i)^T,m_i+\nu_0\right\} \tag{22}$$

4. Simulate $\underline{\mu}_0$ (MATLAB command "mvnrnd")

$$\underline{\mu}_0 \sim \mathrm{N}\left\{\underline{\mu}_0;(\mathbf{C}_{\underline{\mu}_0}^{-1}+M\cdot\mathbf{C}_0^{-1})^{-1}\left(\mathbf{C}_{\underline{\mu}_0}^{-1}\underline{\mu}_{\underline{\mu}_0}+\mathbf{C}_0^{-1}\sum_{i=1}^{M}\underline{\mu}_i\right),(\mathbf{C}_{\underline{\mu}_0}^{-1}+M\cdot\mathbf{C}_0^{-1})^{-1}\right\} \tag{23}$$

where $\underline{\mu}_{\underline{\mu}0}$ = zero vector; and $\mathbf{C}_{\underline{\mu}0}$ = diagonal matrix with diagonals $= 10^4$.

5. Simulate $\mathbf{C}_0$ (MATLAB command "iwishrnd")

$$\mathbf{C}_0 \sim IW\left\{\mathbf{C}_0;\boldsymbol{\Sigma}_{\mathbf{C}_0}+\sum_{i=1}^{M}(\underline{\mu}_i-\underline{\mu}_0)(\underline{\mu}_i-\underline{\mu}_0)^T,M+\nu_{C0}\right\} \tag{24}$$

where $\boldsymbol{\Sigma}_{\mathrm{C0}} = \mathrm{diag}(4/a_1,4/a_2,\ldots,4/a_n)$; and $\nu_{\mathrm{C0}} = n+1$.

6. Simulate $\boldsymbol{\Sigma}_0$ (MATLAB command "wishrnd")

$$\boldsymbol{\Sigma}_0 \sim W\left\{\boldsymbol{\Sigma}_0;\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}^{-1}+\sum_{i=1}^{M}\mathbf{C}_i^{-1},M\nu_0+\lambda_{\boldsymbol{\Sigma}_0}\right\} \tag{25}$$

where $\boldsymbol{\Psi}_{\boldsymbol{\Sigma}_0}$ = diagonal matrix with large diagonals $= 10^4$; and $\lambda_{\boldsymbol{\Sigma}_0} = n+2$.

7. Simulate ν_0 from the following discrete distribution:

$$p(\nu_0|\mathbf{X},\boldsymbol{\mu},\mathbf{C},\underline{\mu}_0,\mathbf{C}_0,\boldsymbol{\Sigma}_0,\underline{a}) \propto \frac{\prod_{i=1}^{M}|\mathbf{C}_i|^{-\frac{\nu_0}{2}}\cdot|\boldsymbol{\Sigma}_0|^{\frac{M\nu_0}{2}}}{2^{Mn\nu_0/2}\cdot\Gamma_n(\nu_0/2)^M}$$

$$\nu_0 = n,n+1,\ldots,1{,}000 \tag{26}$$

8. (For $k=1,\ldots,n$) Simulate a_k [MATLAB command "1/gamrnd(shape,1/scale)"]

$$a_k \sim IG(a_k;\alpha+\frac{n+1}{2},\beta+2/\mathrm{C}_{0,kk}) \tag{27}$$

where $\alpha = 0.5$; and $\beta = 10^{-4}$.

9. (For $i=1,\ldots,M$; $j=1,\ldots,m_i$) Simulate $\underline{x}_{ij}^u$ (MATLAB command "mvnrnd")

$$\underline{x}_{ij}^u \sim N\{\underline{x}_{ij}^u;\underline{\mu}_i^u+\mathbf{C}_i^{uo}(\mathbf{C}_i^o)^{-1}(\underline{x}_{ij}^o-\underline{\mu}_i^o),\mathbf{C}_i^u-\mathbf{C}_i^{uo}(\mathbf{C}_i^o)^{-1}\mathbf{C}_i^{ou}\} \tag{28}$$

10. Cycle Steps 2–9 for T time steps to obtain T samples for $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$.

The GS starts with the initial $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$ and then sequentially draws samples from the conditional PDFs based on the latest parameter values. The samples after the burn-in period (denoted by t_b, determined by visual inspection) are collected. These samples are distributed as $f(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a}|\mathbf{X}^o)$, where $\mathbf{X}^o$ denotes the collection of all observed data in the database. The scatter of the $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$ samples quantifies the statistical uncertainty. If we only keep the samples of the hyperparameters $\boldsymbol{\Theta} = (\underline{\mu}_0,\mathbf{C}_0,\nu_0,\boldsymbol{\Sigma}_0)$, these samples are distributed as $f(\boldsymbol{\Theta}|\mathbf{X}^o)$. They have learned from $\mathbf{X}^o$ (the *observed* transformed soil/rock data) about the hyperparameters $\boldsymbol{\Theta}$ that govern how site-specific parameters $\{(\underline{\mu}_i,\mathbf{C}_i): i=1,\ldots,M\}$ are generated. For the GS, samples at nearby time steps are correlated. To reduce sample correlation and also to reduce subsequent computation, only one $\boldsymbol{\Theta}$ sample is collected for every Δt time steps. At the end, there are $n_{\mathrm{learn}} = (T-t_b)/\Delta t$ samples of $\boldsymbol{\Theta}$, denoted by $\{(\underline{\mu}_{0,t},\mathbf{C}_{0,t},\nu_{0,t},\boldsymbol{\Sigma}_{0,t}):t=t_b+1,t_b+\Delta t+1,\ldots,T\}$, that are distributed as $f(\boldsymbol{\Theta}|\mathbf{X}^o)$.

The aforementioned procedure shows how the hyperparameters $\boldsymbol{\Theta}$ can be simulated when conditioned on the database $\mathbf{X}^o$ containing site-label information. This learning outcome can manifest itself in another way: consider the site-specific parameters $(\underline{\mu}_{M+1},\mathbf{C}_{M+1})$ for an unexplored site, denoted by the $(M+1)$th site. $f(\underline{\mu}_{M+1},\mathbf{C}_{M+1}|\mathbf{X}^o)$ can be sampled based on the $\boldsymbol{\Theta}$ samples:

1. Given each set of $(\underline{\mu}_{0,t},\mathbf{C}_{0,t},\nu_{0,t},\boldsymbol{\Sigma}_{0,t})$ samples, $(\underline{\mu}_{M+1,t},\mathbf{C}_{M+1,t})$ can be simulated as $\underline{\mu}_{M+1,t} \sim N(\underline{\mu}_{M+1};\underline{\mu}_{0,t},\mathbf{C}_{0,t})$ and $\mathbf{C}_{M+1,t} \sim \mathrm{IW}(\mathbf{C}_{M+1};\boldsymbol{\Sigma}_{0,t},\nu_{0,t})$.
2. Do Step 1 for $t=t_b+1,t_b+\Delta t+1,\ldots,T$.

It is correct to say that these $(\underline{\mu}_{M+1},\mathbf{C}_{M+1})$ samples have also learned from $\mathbf{X}^o$. Later in this paper, these $(\underline{\mu}_{M+1},\mathbf{C}_{M+1})$ samples will be referred to as the $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1},\mathbf{C}_{M+1})$ samples.

Computational Benefit for the Multivariate Normal Assumption with Conjugate Priors

The GS can simulate $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$ efficiently by Eqs. (21)–(28) because the current paper adopts the multivariate normal assumption together with conjugate priors such that analytical Bayesian solutions exist for certain conditional probabilities. If the multivariate normality and conjugate priors are not adopted, the simulation of $(\mathbf{X}^u,\boldsymbol{\mu},\mathbf{C},\boldsymbol{\Theta},\underline{a})$ will be computationally challenging because analytical expressions are not available for the conditional PDFs in Eqs. (21)–(28). Instead, one may need to simulate each (scalar)

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component in $(\mathbf{X}^u, \boldsymbol{\mu}, \mathbf{C}, \boldsymbol{\Theta}, \underline{a})$ one by one by Markov chain Monte Carlo (e.g., Metropolis algorithm). In general, the number of components in $(\mathbf{X}^u, \boldsymbol{\mu}, \mathbf{C}, \boldsymbol{\Theta}, \underline{a})$ can be huge: n components in $\underline{\mu}_0$; $n(n+1)/2$ components in $\mathbf{C}_0$; $n(n+1)/2$ components in $\boldsymbol{\Sigma}_0$; one component in ν_0; n components in a; $M \times n$ components in $\boldsymbol{\mu}$; $M \times n(n+1)/2$ components in $\mathbf{C}$; and numerous components in $\mathbf{X}^u$. Moreover, it is required to check whether the simulated $(\mathbf{C}, \mathbf{C}_0, \boldsymbol{\Sigma}_0)$ matrices are positive definite (e.g., through rejection sampling). The problem can easily become computationally infeasible.

The HBM proposed in this paper addresses the computational demand by assuming multivariate normality with conjugate priors. The GS can sample the conditional PDFs in Eqs. (21)–(28) exactly without the need to check the positive definiteness of the $(\mathbf{C}, \mathbf{C}_0, \boldsymbol{\Sigma}_0)$ samples (they are automatically positive definite). Compared with the scenario without multivariate normality and conjugate priors, the saving in the computational cost is tremendous.

Inference Stage: Prediction for a Target Site

Let us consider a target site that is characterized by $(\underline{\mu}_s, \mathbf{C}_s)$. Note that the target site is assumed to be similar, but not identical to, any of the M database sites. Namely, $(\underline{\mu}_s, \mathbf{C}_s)$ are assumed to be govern by the same hyperparameters $\boldsymbol{\Theta}$ that govern the intersite variability implicit in the database. The purpose of the inference stage is to make a prediction by conditioning on both data sets, the generic database $\mathbf{X}$ and the target-site data $\mathbf{X}_s$. Suppose that the target-site data consist of m_s data points: $\mathbf{X}_s = \{\underline{x}_{s,j}: j = 1, \ldots, m_s\}$. There may also be missing values in $\mathbf{X}_s$. Let $\mathbf{X}_s^o$ denote the observed part of $\mathbf{X}_s$ and $\mathbf{X}_s^u$ denote the unobserved part. It is of interest to predict $\mathbf{X}_s^u$ based on all available information, $\mathbf{X}^o$ from the generic database, and $\mathbf{X}_s^o$ from the target site. Namely, it is desirable to sample from $f(\mathbf{X}_s^u|\mathbf{X}^o, \mathbf{X}_s^o)$. Because $\{(\underline{\mu}_{0,t}, \mathbf{C}_{0,t}, \nu_{0,t}, \boldsymbol{\Sigma}_{0,t}): t = t_b + 1, t_b + \Delta t + 1, \ldots, T\}$ are distributed as $f(\boldsymbol{\Theta}|\mathbf{X}^o)$, it follows that $f(\mathbf{X}_s^u|\mathbf{X}^o, \mathbf{X}_s^o)$ can be sampled using the following procedure:

1. Given each set of $(\underline{\mu}_{0,t}, \mathbf{C}_{0,t}, \nu_{0,t}, \boldsymbol{\Sigma}_{0,t})$ samples, $f(\mathbf{X}_s^u|\underline{\mu}_{0,t}, \mathbf{C}_{0,t}, \nu_{0,t}, \boldsymbol{\Sigma}_{0,t}, \mathbf{X}_s^o)$ is sampled by the following GS:
   - Initialize $(\mathbf{X}_s^u, \underline{\mu}_s, \mathbf{C}_s)$ samples at arbitrary values. Note that $\mathbf{X}_s$ does not contain missing values after the initialization.
   - Simulate $\underline{\mu}_s$ (MATLAB command "mvnrnd")

$$\underline{\mu}_s \sim \mathrm{N}\left\{\underline{\mu}_s; (\mathbf{C}_{0,t}^{-1} + m_s \mathbf{C}_s{}^{-1})^{-1}\left(\mathbf{C}_{0,t}^{-1}\underline{\mu}_{0,t} + \mathbf{C}_s^{-1}\sum_{j=1}^{m_s}\underline{x}_{s,j}\right), (\mathbf{C}_{0,t}^{-1} + m_s\mathbf{C}_s^{-1})^{-1}\right\} \tag{29}$$

   - Simulate $\mathbf{C}_s$ (MATLAB command "iwishrnd")

$$\mathbf{C}_s \sim \mathrm{IW}\left\{\mathbf{C}_s; \boldsymbol{\Sigma}_{0,t} + \sum_{j=1}^{m_s}(\underline{x}_{s,j} - \underline{\mu}_s)(\underline{x}_{s,j} - \underline{\mu}_s)^T, m_s + \nu_{0,t}\right\} \tag{30}$$

   - (For $j = 1, \ldots, m_s$) Simulate $\underline{x}_{s,j}^u$ (MATLAB command "mvnrnd")

$$\underline{x}_{s,j}^u \sim \mathrm{N}\{\underline{x}_{s,j}^u; \underline{\mu}_s^u + \mathbf{C}_s^{uo}(\mathbf{C}_s^o)^{-1}(\underline{x}_{s,j}^o - \underline{\mu}_s^o), \mathbf{C}_s^u - \mathbf{C}_s^{uo}(\mathbf{C}_s^o)^{-1}\mathbf{C}_s^{ou}\} \tag{31}$$

   - Cycle 2nd to 4th items above for T' time steps to obtain T' samples for $(\mathbf{X}_s^u, \underline{\mu}_s, \mathbf{C}_s)$. The burn-in period, denoted by t_b', is determined by visual inspection. To reduce sample correlation, one can also extract one sample for every $\Delta t'$ time steps. At the end, there are $n_{\mathrm{inf}} = (T' - t_b')/\Delta t'$ samples of $\mathbf{X}_s^u$, $\{\mathbf{X}_{s,t'}^u: t' = t_b' + 1, t_b' + \Delta t' + 1, \ldots, T'\}$, that are distributed as $f(\mathbf{X}_s^u|\underline{\mu}_{0,t}, \mathbf{C}_{0,t}, \nu_{0,t}, \boldsymbol{\Sigma}_{0,t}, \mathbf{X}_s^o)$. Note that we have adopted the prime notations (t', T', t_b') for the GS in the inference stage. The nonprime notations (t, T, t_b) are reserved for the GS in the learning stage.
2. Do Step 1 for $t = t_b + 1, t_b + \Delta t + 1, \ldots, T$. The resulting $n_{\mathrm{learn}} \times n_{\mathrm{inf}}$ samples of $\mathbf{X}_s^u$ are distributed as $f(\mathbf{X}_s^u|\mathbf{X}^o, \mathbf{X}_s^o)$. These $n_{\mathrm{learn}} \times n_{\mathrm{inf}}\,\mathbf{X}_s^u$ samples are converted to the physical soil parameters $\mathbf{Y}_s^u$ through Eq. (2). The $\mathbf{Y}_s^u$ samples can be used to predict the statistics of the soil/rock parameters for the target site, e.g., their conditional median values and 95% CIs.

It is useful to note that there are two Gibbs samplers in the entire procedure, one in the learning stage and one in the inference stage. The purpose for the GS in the learning stage is to obtain the hyperparameter samples $\{(\underline{\mu}_{0,t}, \mathbf{C}_{0,t}, \nu_{0,t}, \boldsymbol{\Sigma}_{0,t}): t = t_b + 1, t_b + \Delta t + 1, \ldots, T\}$ based on the observed transformed soil/rock data $\mathbf{X}^o$ containing site-label information, whereas the purpose for the GS in the inference stage is to predict the missing values for the target site $\mathbf{X}_s^u$ based on the hyperparameter samples. If the inference for another target site is needed, the same hyperparameter samples obtained in the learning stage can be used. There is no need to redo the learning stage, and it is only necessary to conduct the inference stage for the new target site.

Numerical Examples

To demonstrate the effectiveness of the proposed HBM and GS algorithms, the following numerical example is considered. Suppose the database $\mathbf{X}^o$ consists of (x_1, x_2) data points from 100 sites ($M = 100$), and each site contributes 10 data points ($m_1 = m_2 = \ldots = m_{100} = 10$). As a result, there are in total 1,000 (x_1, x_2) data points in $\mathbf{X}^o$. The data points for the ith site in the database are generated using the following procedure:

1. Generate $\underline{\mu}_i \sim N(\underline{\mu}_i; \underline{\mu}_0, \mathbf{C}_0)$, where $\underline{\mu}_0 = [0; 0]$ and $\mathbf{C}_0 =$ [1 0; 0 1].
2. Generate $\mathbf{C}_i \sim \mathrm{IW}(\mathbf{C}_i; \boldsymbol{\Sigma}_0, \nu_0)$, where $\boldsymbol{\Sigma}_0 = 5\times$ [1 0.9; 0.9 1] and $\nu_0 = 8$. Because the mean value of $\mathbf{C}_i$ is equal to $\boldsymbol{\Sigma}_0/(\nu_0 - n - 1)$, $\mathbf{C}_i$ has a mean equal to [1 0.9; 0.9 1]. The simulated $\mathbf{C}_i$ can be expressed as the following form:

$$\mathbf{C}_i = \begin{bmatrix} \sigma_{i,1}^2 & \rho_i\sigma_{i,1}\sigma_{i,2} \\ \rho_i\sigma_{i,1}\sigma_{i,2} & \sigma_{i,2}^2 \end{bmatrix} \tag{32}$$

   where $\sigma_{i,1}$ and $\sigma_{i,2}$ = site-specific standard deviations for the ith site; and ρ_i = site-specific correlation.
3. For the jth data point ($j = 1, \ldots, 10$), generate $\underline{x}_{ij} \sim N(\underline{x}_{ij}; \underline{\mu}_i, \mathbf{C}_i)$.

Fig. 6 shows the generated site-specific parameters, including the site-specific means $(\underline{\mu}_{i,1}, \underline{\mu}_{i,2})$, site-specific standard deviations $(\sigma_{i,1}, \sigma_{i,2})$, and the site-specific correlation ρ_i. Fig. 7 shows the resulting (x_1, x_2) data points, denoted by $\mathbf{X}^o = \{\underline{x}_{ij}: i = 1, \ldots, 100, j = 1, \ldots, 10\}$. The data points for one particular site are shown as larger circles. For this particular site, the site-specific mean is marked as + in the figure. The data points for the remaining 99 sites are shown as smaller dots. Note that the hyperparameters $\boldsymbol{\Theta} = (\underline{\mu}_0, \mathbf{C}_0, \nu_0, \boldsymbol{\Sigma}_0)$ and generated site-specific normal model parameters $(\underline{\mu}_i, \mathbf{C}_i)$ are known during the data generation only, and they are treated as unknown during the learning and inference stages.

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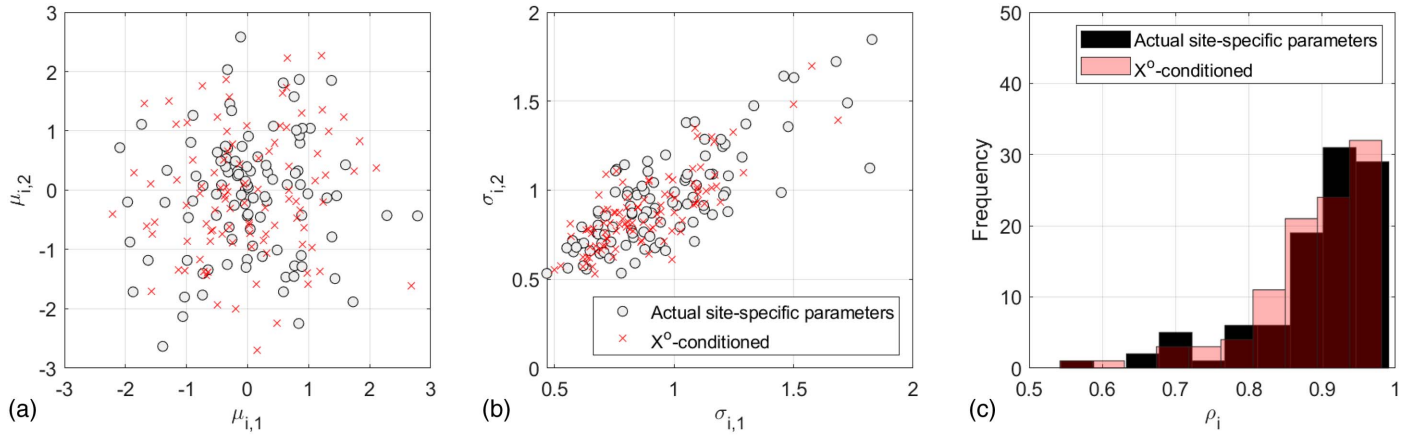


Fig. 6. Site-specific parameters for the numerical example: (a) $\underline{\mu}_i$; (b) $\sigma_{i,1}$ and $\sigma_{i,2}$; and (c) ρ_i.

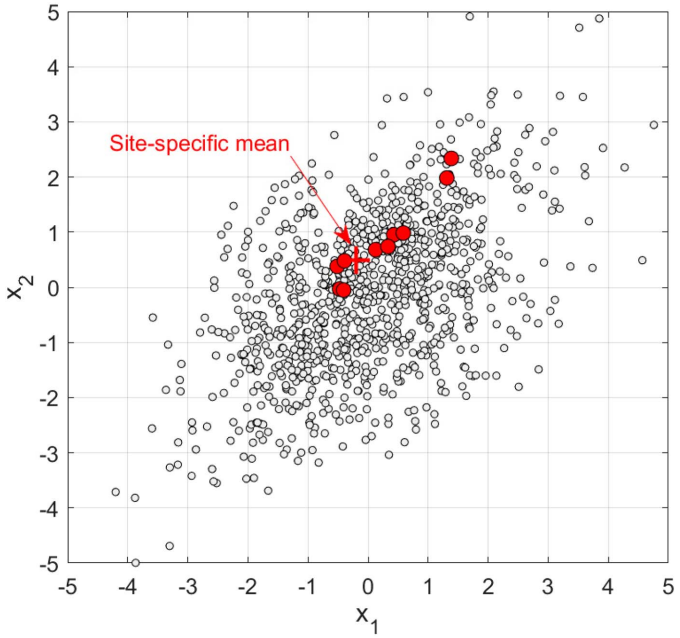


Fig. 7. Simulated (x_1, x_2) data for the database $\mathbf{X}^o$ (numerical example).

Learning Stage

Given the database $\mathbf{X}^o$, the GS is adopted to sample the hyperparameters $\boldsymbol{\Theta}$. The total time step T for the GS is taken to be 21,000, and the end of burn-in period is determined to be $t_b = 1{,}000$ by visual inspection of the time history of the GS samples. To reduce sample correlation and also to reduce subsequent computation, only one $\boldsymbol{\Theta}$ sample is collected for every $\Delta t = 10$ time steps. At the end, there are $n_{\text{learn}} = (T - t_b)/\Delta t = 2{,}000$ samples of $\boldsymbol{\Theta}$. Fig. 8 shows the scatter plots and histograms for these 2,000 samples. In the figure, $\underline{\mu}_{0,i}$ denotes the ith entry in the $\underline{\mu}_0$ vector, and $C_{0,ij}$ denotes the (i, j) entry in the $\mathbf{C}_0$ matrix. The cross marks and vertical lines in the figure indicate the underlying actual values of the hyperparameters (treated as unknown during the GS). It is remarkable that the GS samples are consistent with the underlying actual values, suggesting that the GS can consistently learn from $\mathbf{X}^o$. The consistency for the GS learning outcome can also be seen by comparing the $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1}, \mathbf{C}_{M+1})$ samples (the cross marks in Fig. 6) with the underlying site-specific parameters for the 100 sites in the database.

Inference Stage

Now consider a target site with 10 data points ($m_s = 10$). The target-site parameters $\underline{\mu}_s = [-0.44; 0.85]$ and $\mathbf{C}_s = [1.77\ 1.85; 1.85\ 2.10]$ are also generated based on the same underlying hyperparameters $\boldsymbol{\Theta}$. Then, the target-site data $\mathbf{X}_s$ are generated based on $\underline{\mu}_s$ and $\mathbf{C}_s$. Among the 10 data points, three data points are with complete (x_1, x_2) information, and seven data points are with x_1 values only (x_2 values are missing). Here, x_1 mimics an inexpensive soil/rock parameter (e.g., CPT cone tip resistance of a clay), whereas x_2 mimics an expensive soil/rock parameter (e.g., undrained shear strength of a clay). The purpose of the inference is to predict these missing x_2 values. Both the target-site parameters $(\underline{\mu}_s, \mathbf{C}_s)$ and the missing x_2 values are treated as unknown during the inference stage. Fig. 9(a) shows the target-site data as circles. The dark circles are the target-site data with complete (x_1, x_2) information, whereas the light circles are the target-site data with missing x_2 values. The dashed ellipse in Fig. 9(a) indicates the 95% confidence ellipse (CE) based on the underlying $(\underline{\mu}_s, \mathbf{C}_s)$. For comparison, the data points for the generic database are shown in Fig. 9(a) as grey dots. For the GS during the inference stage, $t_b' = 20$, $T' = 21$, and $\Delta t' = 1$, i.e., only one $\mathbf{X}_s^u$ sample is obtained for each GS ($n_{\text{inf}} = 1$). At the end, there are $n_{\text{learn}} \times n_{\text{inf}} = 2{,}000$ samples of $\mathbf{X}_s^u$.

Let us consider the following two scenarios for the inference:

1. The inference is for data points with missing x_2 [light circles in Fig. 9(a)]. The 2,000 $\mathbf{X}_s^u$ samples can be used to infer these missing x_2 values. The vertical line segments in Fig. 9(a) show the 95% CI for the missing x_2 values. It is remarkable that all 95% CIs contain the actual x_2 values, which are treated as unknown during the inference stage. Fig. 9(b) further compares the median estimates of the missing x_2 values with their actual values. It is clear that the median estimates are close to their actual values.
2. The inference is for a new data point at the target site with unknown x_1 and x_2. This can be done by augmenting one empty row (both x_1 and x_2 are missing) into $\mathbf{X}_s$, and these missing x_1 and x_2 values are also sampled. The darker dots in Fig. 9(a) show the (x_1, x_2) samples. It is remarkable that the (x_1, x_2) samples can capture the underlying 95% CE (treated as unknown) quite well.

Comparison with Hybridization

The hybridization method proposed by Ching and Phoon (2019) is adopted to analyze this numerical example. The generic data points

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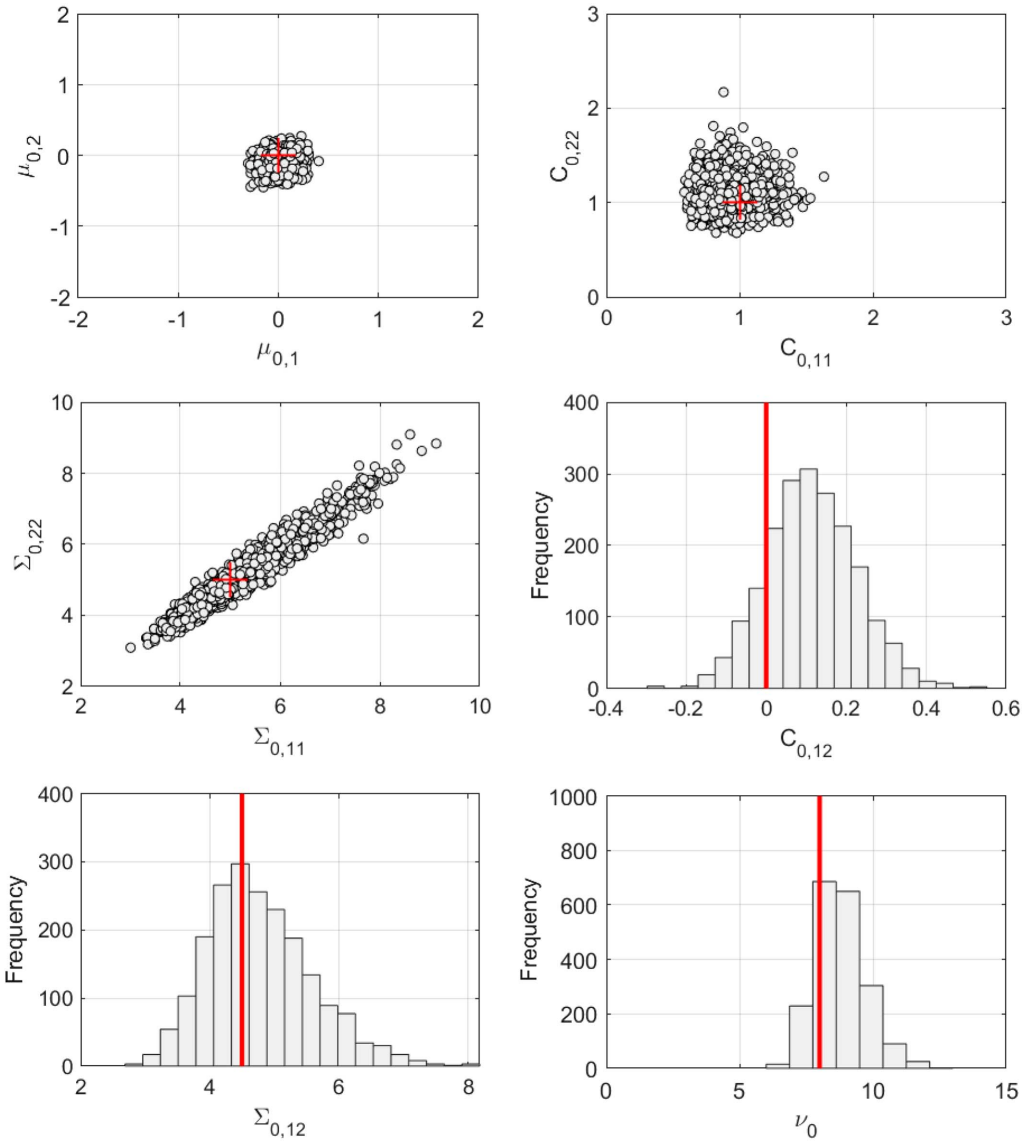


Fig. 8. Scatter plots and histograms for the GS samples of $(\underline{\mu}_0, \mathbf{C}_0, \nu_0, \mathbf{\Sigma}_0)$.

$\mathbf{X}^o$ are analyzed to obtain the generic PDF [Fig. 10(a) depicts its contour plot]. Then, the target-site data $\mathbf{X}_s^o$ are analyzed to obtain the target-site PDF [Fig. 10(b) depicts its contour plot], which is very wide in the x_2 direction. This is because there are only three data points with known x_2, so the statistical uncertainty in x_2 is large. Fig. 10(c) depicts the contour plot for the hybrid PDF.

Based on the hybrid PDF, the x_2 value for data points with missing x_2 can be sampled (Scenario 1). The vertical line segments in Fig. 11(a) show the 95% CIs for the x_2 values. Although all 95% CIs contain the actual x_2 values, they are significantly wider than those in Fig. 9(a). Also, the (x_1, x_2) values at a new data point can be sampled (Scenario 2). The dots in Fig. 11(a) show the (x_1, x_2) samples. Note that they do not capture the underlying 95% CE for the target site well. Fig. 11(b) further compares the median estimates of the missing x_2 values with their actual values. It is clear that the median estimates somewhat deviate from their actual values. These observations suggest that the HBM (Fig. 9) is more effective than the hybridization method (Fig. 11) for the current example.

The aforementioned numerical example is for the case with seven deliberately missing x_2 values. It is interesting to see how the conclusion varies with the number of deliberately missing x_2 values. Let us consider four cases with different numbers of deliberately missing x_2 values:

1. (Case 1) 10 x_2 values are missing (zero observed).
2. (Case 2) Nine x_2 values are missing (one observed).
3. (Case 3) Seven x_2 values are missing (three observed). This is the previous case.
4. (Case 4) Five x_2 values are missing (five observed).

For all cases, a new data point with missing x_1 and x_2 is considered by augmenting one empty row into $\mathbf{X}_s$. Fig. 12 shows the (x_1, x_2) samples based on the HBM, whereas Fig. 13 shows those based on the hybridization method. For Case 1, both HBM and hybridization fail to capture the underlying 95% CE for the target site (dashed ellipse). As the amount of x_2 information increases, the HBM quickly captures the underlying 95% CE (Fig. 12), whereas the hybridization method can only slowly capture the underlying 95% CE (Fig. 13). The median estimates made by the HBM for the missing x_2 values are also generally more accurate than those made by the hybridization method. Because of its hierarchical structure (clusters of distributions), the HBM is more efficient in extracting the site-label information than the hybridization method. This result

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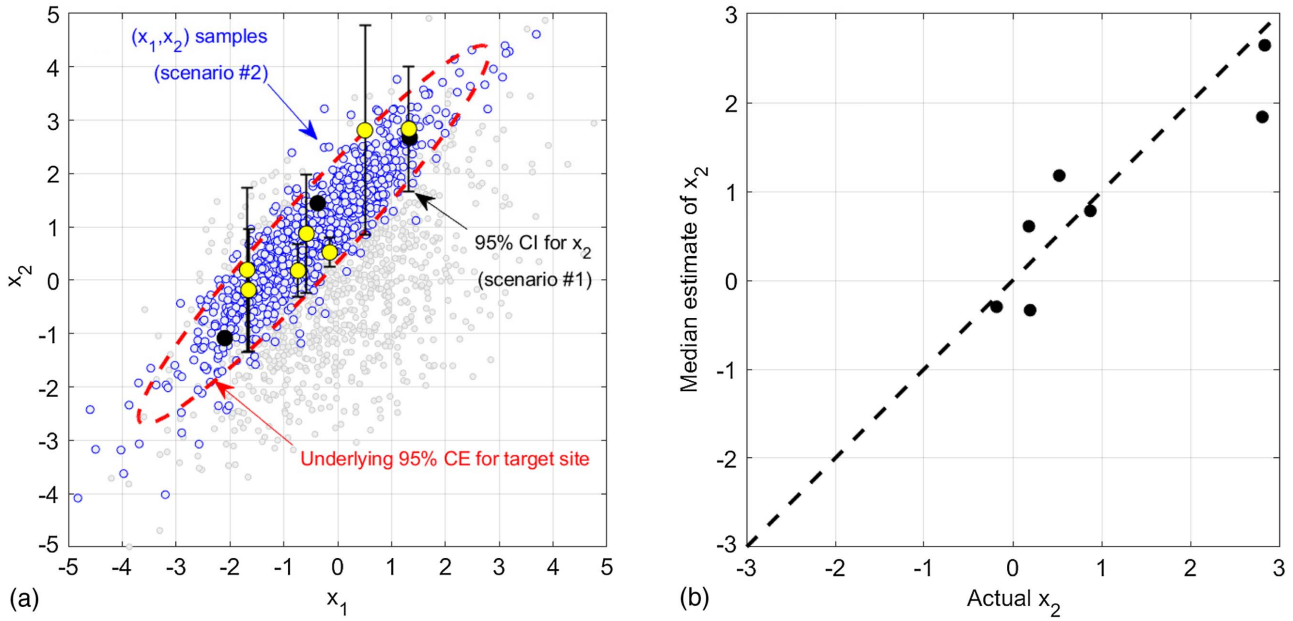


Fig. 9. Inference results for the numerical example based on HBM: (a) 95% CIs for x_2; and (b) median estimate for x_2.

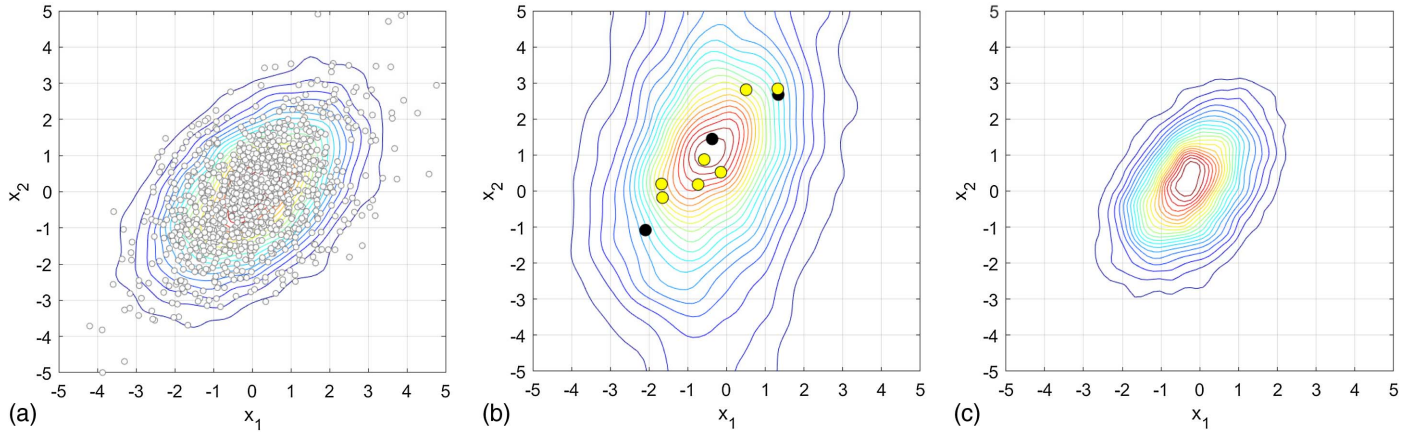


Fig. 10. (a) Generic PDF constructed based on $\mathbf{X}^o$; (b) target-site PDF constructed based on $\mathbf{X}_s^o$; and (c) hybrid model.

is somewhat to be expected since this simulated data also have the hierarchical structure.

Real Example

Generic Database

For this real example, consider the y_1 versus y_2 data points in CLAY/10/7490 [recall that $y_1 = \ln[(q_t - \sigma_v)/\sigma'_v]$ and $y_2 = \ln(s_u/\sigma'_v)$]. The y_1–y_2 data points from 91 sites are extracted to form the generic database $\mathbf{X}^o$. The number of data points for each site ranges from 3 to 38 (on average 12). Although there are missing values in the database (either y_1 or y_2 is missing), 81% of the data points are with complete y_1–y_2 information. Some of the data points with complete y_1–y_2 information are shown in Fig. 3(a), where the data points with the same symbol are from the same site location. The (y_1, y_2) data are converted to the (x_1, x_2) data by Eq. (2). The site-specific statistics (sample means, sample standard deviations, and sample correlation) for each of the 91 sites are estimated based on the site-specific (x_1, x_2) data. These site-specific statistics are shown in Fig. 14 as circles.

Learning Stage

Given the database $\mathbf{X}^o$, the GS is adopted to learn the hyperparameters $\boldsymbol{\Theta}$ in the learning stage. The total time step T is taken to be 21,000, the end of burn-in period is determined to be $t_b = 1{,}000$, and one $\boldsymbol{\Theta}$ sample is collected for every 10 time steps ($\Delta t = 10$). As a result, there are $n_{\text{learn}} = (T - t_b)/\Delta t = 2{,}000$ samples. These $\boldsymbol{\Theta}$ samples are used to simulate $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1}, \mathbf{C}_{M+1})$ samples. Fig. 14 shows the site-specific statistics (means, standard deviations, and correlation) based on the $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1}, \mathbf{C}_{M+1})$ samples as the cross marks and histogram. They are similar to the actual statistics of the 91 sites.

Inference Stage

Now consider a Taipei (Taiwan) site as the target site. The data points for this Taipei site are not in the generic database $\mathbf{X}^o$. Table 2 gives

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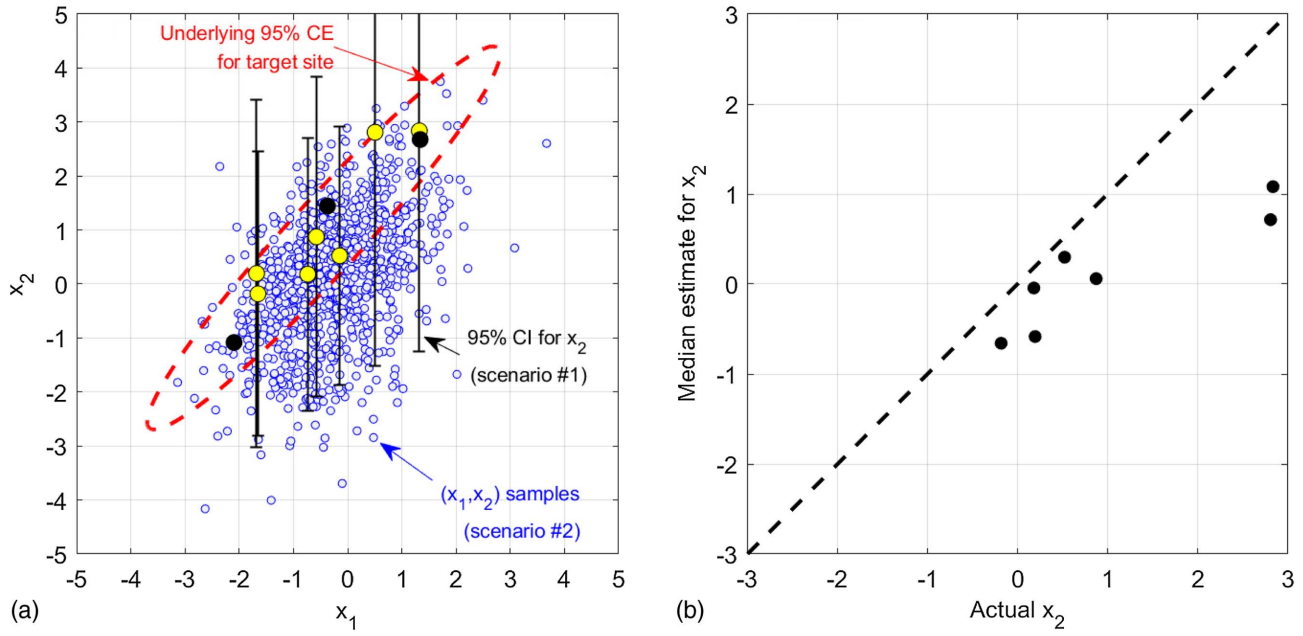


Fig. 11. Inference results for the numerical example based on the hybridization method: (a) 95% CIs for x_2; and (b) median estimate for x_2.

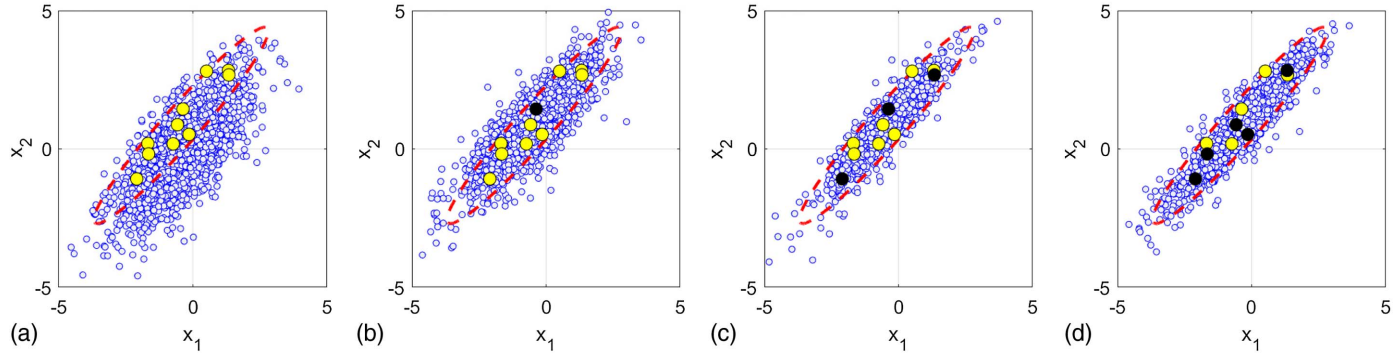


Fig. 12. (x_1, x_2) samples based on HBM: (a) Case 1; (b) Case 2; (c) Case 3; and (d) Case 4.

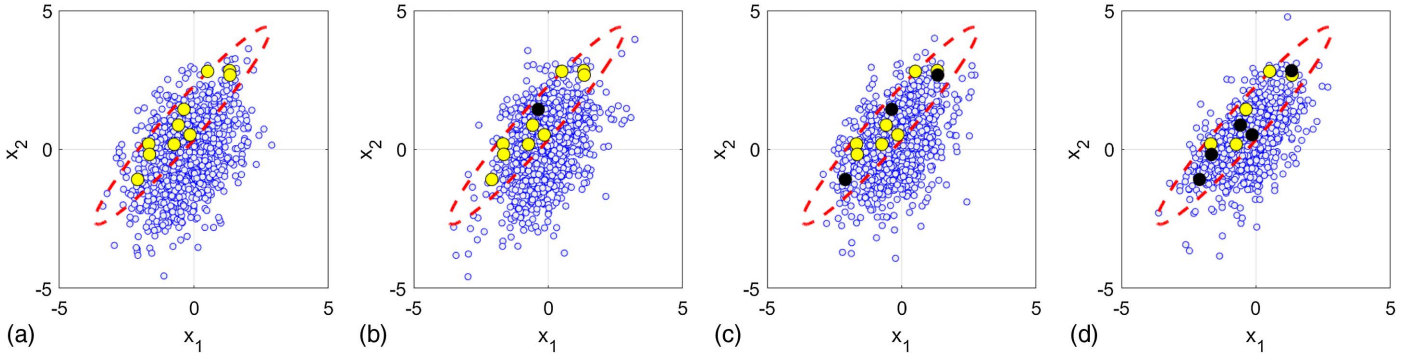


Fig. 13. (x_1, x_2) samples based on the hybridization method: (a) Case 1; (b) Case 2; (c) Case 3; and (d) Case 4.

the site investigation results for a silty clay layer at this Taipei site (Ou and Liao 1987). For this site, three boreholes and three CPT soundings are conducted. The data in Table 2 are based on one borehole and one nearby CPT sounding. For the time being, only the (y_1, y_2) data in Table 2 are considered in the following analysis. The (y_1, y_2) data are also converted to (x_1, x_2) by Eq. (2) to form the target-site data $\mathbf{X}_s^o$.

To demonstrate the inference results, the x_2 values (hence s_u/σ_v') at some depths are deliberately treated as missing. The purpose of the inference is to infer these missing x_2 values. In the

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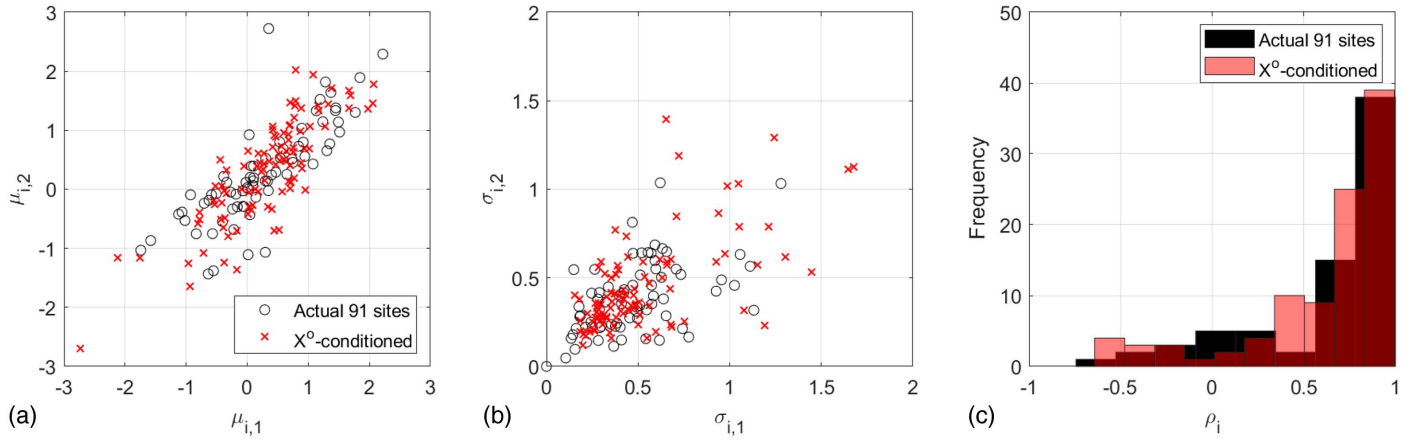


Fig. 14. Site-specific statistics for the generic database: (a) $\underline{\mu}_i$; (b) $\sigma_{i,1}$ and $\sigma_{i,2}$; and (c) ρ_i.

Table 2. Site investigation results for a silty clay layer at a Taipei site

Depth (m)	LL	PI	LI	σ'_v (kPa)	σ'_p (kPa)	$(q_t-\sigma_v)/\sigma'_v$ [exp(y_1)]	s_u/σ'_v [exp(y_2)]	Four cases for s_u/σ'_v [exp(y_2)]			
								#1	#2	#3	#4
12.8	30.1	9.1	1.20	128.0	172.7	5.17	0.37	—	—	0.37	0.37
14.8	32.8	12.8	1.43	144.9	—	4.22	0.36	—	—	—	—
16.1	36.4	14.5	1.24	155.6	—	4.12	0.33	—	—	—	0.33
17.8	41.9	18.9	0.90	169.9	181.8	4.03	0.25	—	—	—	—
18.3	—	—	—	174.5	—	5.27	0.34	—	0.34	0.34	0.34
20.2	38.1	17.3	0.70	190.0	—	4.53	0.32	—	—	—	—
22.7	37.0	16.0	0.58	210.9	—	4.76	0.31	—	—	—	0.31
24.0	38.0	16.2	0.75	221.7	221.7	5.12	0.30	—	—	—	—
26.6	34.8	13.8	0.80	243.7	—	5.32	0.34	—	—	0.34	0.34

current demonstration, there is only one target site (Taipei site), but the HBM can predict properties at m sites ($m \geq 1$) simultaneously. Also, in the current demonstration, the prediction target is x_2 (hence s_u/σ'_v), but the HBM can predict other variables as well (such as preconsolidation stress σ'_p) as long as they are covered by the CLAY/10/7490 database. The HBM can also predict the multivariate PDF of multiple parameters simultaneously, e.g., the posterior bivariate PDF of (s_u, σ'_p). Four cases with different numbers of deliberately missing x_2 values are considered (see the right columns in Table 2):

1. (Case 1) nine x_2 values are missing (zero observed).
2. (Case 2) eight x_2 values are missing (one observed at depth = 18.3 m).
3. (Case 3) six x_2 values are missing (three observed at depths = 12.8, 18.3, and 26.6 m).
4. (Case 4) four x_2 values are missing (five observed at depths = 12.8, 16.1, 18.3, 22.7, and 26.6 m).

The purpose of the inference is to estimate the missing x_2.

For the GS in the inference stage, $T' = 120$, $t'_b = 20$, and $\Delta t' = 10$ are adopted, i.e., $n_{\text{inf}} = (T' - t'_b)/\Delta t' = 10$. At the end, there are $n_{\text{learn}} \times n_{\text{inf}} = 20{,}000$ samples of $\mathbf{X}^u_s$. The solid dark line segments in Fig. 15 show the 95% CIs of the undrained shear strength (s_u) at the depths where x_2 is deliberately treated as missing (yellow circles). The dark circles in the figure are for the depths with complete information, so the inference is not necessary. Recall that $y_2 = \ln(s_u/\sigma'_v)$, so the upper and lower bounds for the 95% CI of y_2 are converted to the upper and lower bounds for s_u by making use of the relation $s_u = \exp(y_2) \times \sigma'_v$. It is remarkable that the 95% CIs contain the actual s_u values (light circles in the figure), which are treated as unknown during the analysis. The dotted dark line segments in Fig. 15 show the 95% CIs obtained by the hybridization method, and the grey line segments show the 95% CIs obtained by Ching and Phoon's (2014a) method (the 95% CIs in Fig. 4). Ching and Phoon's (2014a) method constructs the generic PDF $f(y_1, y_2)$ based on the CLAY/10/7490 database. This $f(y_1, y_2)$ is then treated as the prior PDF for the target (Taipei) site. The posterior PDF of y_2, namely, $f(y_2|y_1)$, can be obtained by conditioning on target-site y_1 information. The posterior median and 95% CI for $f(y_2|y_1)$ are shown in Fig. 4(a). This method provides a set of probabilistic regression lines (the posterior median and 95% CI) and hence is termed *probabilistic multiple regression* from here on. Note that the probabilistic multiple regression does not distinguish between the local (target-site) and generic (database) populations. Moreover, once the probabilistic multiple regression model in Fig. 4(a) is developed, it is fixed and cannot be updated by the target-site data.

Among the three methods (HBM, hybridization, and probabilistic multiple regression), the probabilistic multiple regression gives the widest 95% CI in Fig. 15, hence the largest transformation uncertainty. The performance of the HBM and hybridization methods are comparable (the widths of 95% CIs are similar) when the target-site data are sparse (Case 1) or when the target-site data are abundant (Case 4). However, when the amount of the target-site data is intermediate (Cases 2 and 3), the HBM is more effective than the hybridization method because the HBM gives narrower 95% CIs. The superior performance of the HBM might be due to the fact that the HBM can extract the site-label information in the CLAY/10/7490 database, whereas the hybridization method cannot.

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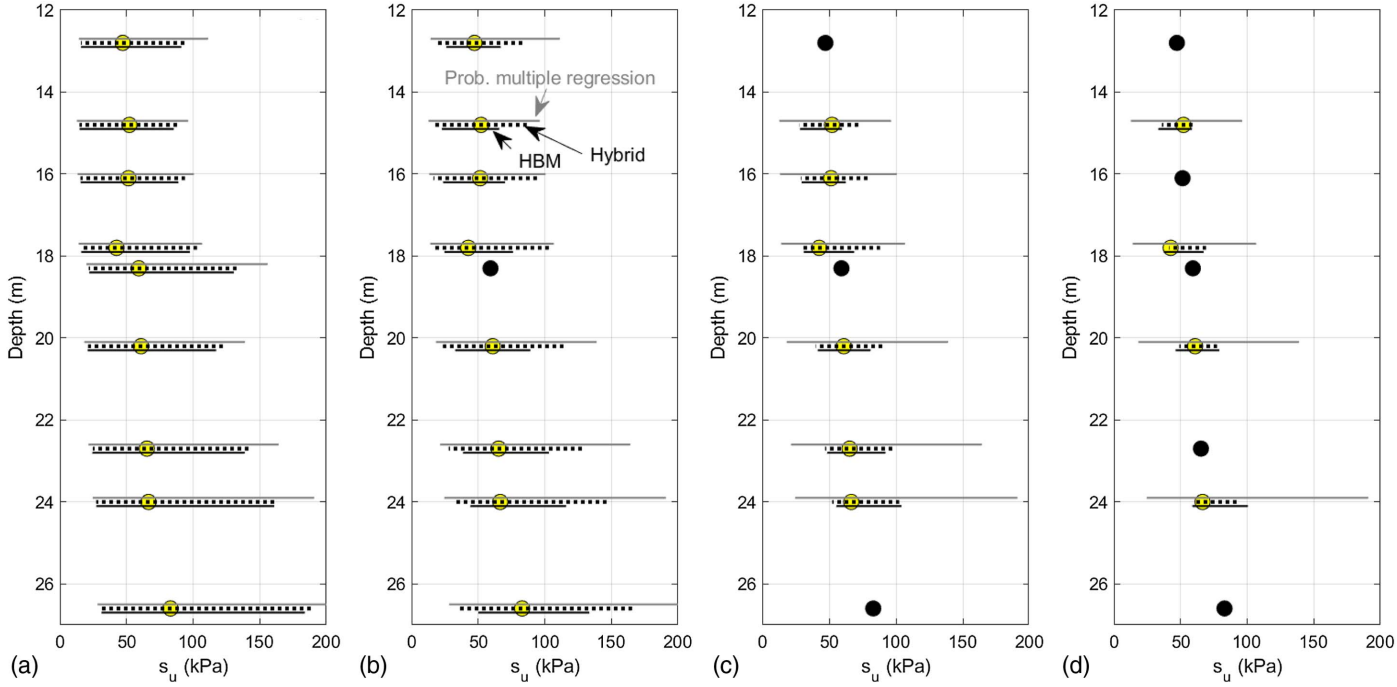


Fig. 15. Inference results for (a) Case 1, (b) Case 2, (c) Case 3; and (d) Case 4 for the real example.

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Multivariate Learning and Inference

There are seven soil parameters [liquid limit (LL), plasticity index (PI), liquidity index (LI), σ'_v, σ'_p, $(q_t - \sigma'_v)/\sigma'_v$, and s_u/σ'_v] for the Taipei-site data in Table 2. Figs. 14 and 15 only show the learning and inference results for bivariate inference [only $(q_t - \sigma_v)/\sigma'_v$ and s_u/σ'_v are considered]. In this section, all seven parameters are considered, so the problem dimension becomes $n = 7$. All parameters become high dimensional, e.g., $\underline{\mu}_i \in \mathbf{R}^{7\times 1}$ and $\mathbf{C}_i \in \mathbf{R}^{7\times 7}$. Nonetheless, the proposed HBM and GS algorithm are still applicable. For the learning stage, 4,766 data points from 340 sites are extracted from CLAY/10/7490. Nearly all data points, except for a small fraction, are with incomplete information (only a subset of the seven parameters are known). The $(y_1, y_2, \ldots, y_7)$ values can be converted to $(x_1, x_2, \ldots, x_7)$ through equations similar to Eq. (2) (see Ching and Phoon 2014a for details). For the GS in the learning stage, $T = 21{,}000$, $t_b = 1{,}000$, and $\Delta t = 10$, so there are $n_{\text{learn}} = (T - t_b)/\Delta t = 2{,}000$ samples of $\boldsymbol{\Theta}$. The detailed results of these $\boldsymbol{\Theta}$ samples are not presented for brevity. The 2,000 $\boldsymbol{\Theta}$ samples are used to simulate $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1}, \mathbf{C}_{M+1})$ samples. Fig. 16 shows the histograms for the site-specific correlation coefficients based on the actual data from the 340 sites versus those based on the $\mathbf{X}^o$-conditioned $(\underline{\mu}_{M+1}, \mathbf{C}_{M+1})$ samples. General consistency is found.

For the inference stage, let us consider the Taipei-site data in Table 2. Again, the s_u/σ'_v values at some depths are deliberately treated as missing (the same four cases as the right columns in Table 2), and the purpose is to estimate the s_u values at these depths, but now not only based on the information of $(q_t - \sigma_v)/\sigma'_v$ but also based on the information of LL, PI, LI, σ'_v, and σ'_p. For the GS in the inference stage, $T' = 200$, $t'_b = 100$, and $\Delta t' = 10$ are adopted, i.e., $n_{\text{inf}} = (T' - t'_b)/\Delta t' = 10$. At the end, there are $n_{\text{learn}} \times n_{\text{inf}} = 20{,}000$ samples of $\mathbf{X}^u_s$. Fig. 17 shows the inference results in terms of the 95% CIs of s_u. Compared with Fig. 15, some of the 95% CIs become narrower, probably because of the availability of multivariate information. The HBM tends to be more effective than the hybridization method because the HBM gives narrower 95% CIs. There is an exception: for Case 4, the hybridization method gives narrower CIs, but one of its CIs does not contain the actual s_u value (depth = 17.8 m).

Computer Runtimes

The computer runtimes of the real examples (for both learning and inference) for the HBM and hybridization methods are compared in Table 3. The runtimes are based on MATLAB codes in a personal computer with Intel Core i9-9920X clocked at 3.5GHz and 64GB RAM. The bivariate cases in the table refer to those with $n = 2$ (i.e., Fig. 15), whereas the multivariate cases refer to those with $n = 7$ (i.e., Fig. 17). The runtimes for Cases 1–4 are roughly equal, so Table 3 shows the average runtimes among the four cases. It is evident that the HBM requires 20%–40% more runtime than the hybridization method. For both methods, the runtimes seem to scale linearly with the number of soil parameters (n). To investigate how the runtime scales with the size of the generic database ($\mathbf{X}^o$), the HBM and hybridization are executed with $n = 7$, but the number of records in $\mathbf{X}^o$ is reduced to about 54%. The runtimes for this reduced $\mathbf{X}^o$ are shown in the rightmost column in Table 3. Compared with the runtimes for full $\mathbf{X}^o$, the runtimes for reduced $\mathbf{X}^o$ seem to scale linearly with the number of records in $\mathbf{X}^o$. It is worth noting that the computations for the real examples can be finished in minutes because of the assumed multivariate normality and conjugate priors. If the multivariate normality assumption and conjugate priors are not adopted, the runtime can be very long.

Conclusion

An efficient HBM with closed-form conditional probabilities is proposed in the current paper to construct a quasi-site-specific multivariate PDF for soil/rock properties based on the site-specific data and a generic soil/rock database. This HBM with closed-form conditional probabilities is crucial to manage computational

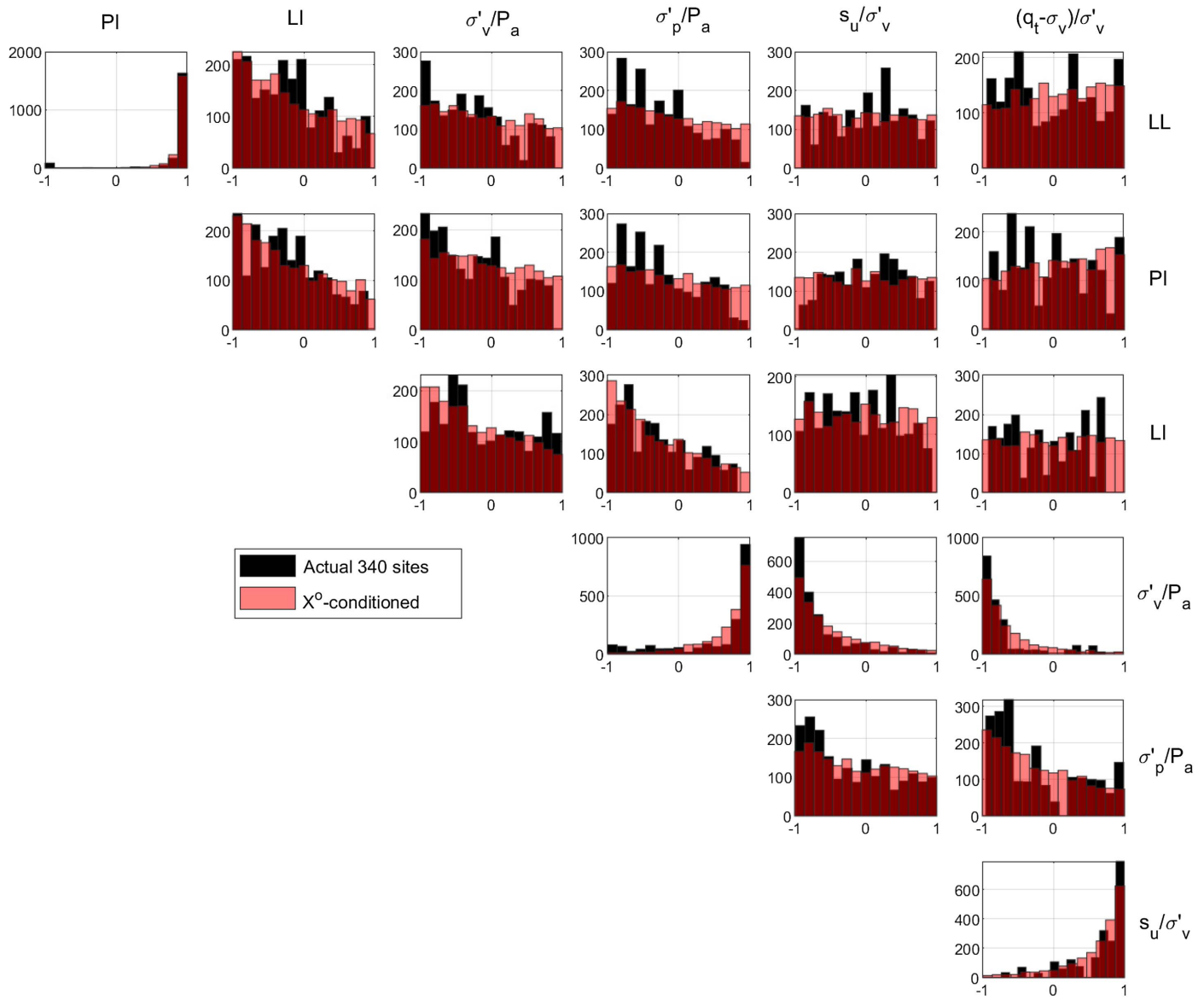


Fig. 16. Site-specific correlation coefficients for the multivariate case.

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demand for actual large geotechnical databases. The focus of this paper is to study the existence of groups in a generic soil/rock database. Data within a group are more similar to each other than to data in other groups. Grouping criterion can be based on site localization, geologic origin, soil/rock types, and others. This paper only considers site localization in the qualitative sense of data measured in a single project site as belonging to one group. In short, the generic database is further divided into data groups labeled by qualitative site labels. No other site attributes such as size of the site, GPS location, region, country, or others are considered. Hence, this site label does not contain any quantitative information, beyond the qualitative information that each group is distinct. This grouping criterion is useful, because it relies on marginally more information than treating the generic database as a single group. It also addresses site uniqueness, which is a central issue in geotechnical engineering. It is worth mentioning that a site label does not reveal the location of a site, which can be confidential. This data anonymization aspect is critical in practice (Phoon et al. 2021).

The HBM has a special model structure with hyperparameters that govern the intersite mean and intersite variability in the generic soil database. Hence, it allows groups (sites) to be distinct populations, but requires their distinctiveness to be distributed according to the hyperparameters. With this special model structure, it is shown that the HBM can effectively extract the site-label information in the generic soil database, whereas an existing hybridization method cannot. It is also shown that the HBM can produce a more effective quasi-site-specific multivariate PDF than the hybridization method. It is important to note that while the existence of groups based on site label or geology is obvious from physical considerations, the ability to identify such groups statistically from data alone has not been systematically studied thus far based on an actual large generic database such as CLAY/10/7490.

This paper assumes that the heterogeneity between groups can be modeled using the HBM and achieves some success for the grouping criterion based on site labels. This may justify the use of the HBM hyperparameters to capture intersite heterogeneity, but more research is needed. The challenge of modeling site differences in an actual soil/rock database that can lead to better local transformation models (the site challenge) remains open, but this paper takes a reasonable first stab at this challenge while satisfying two

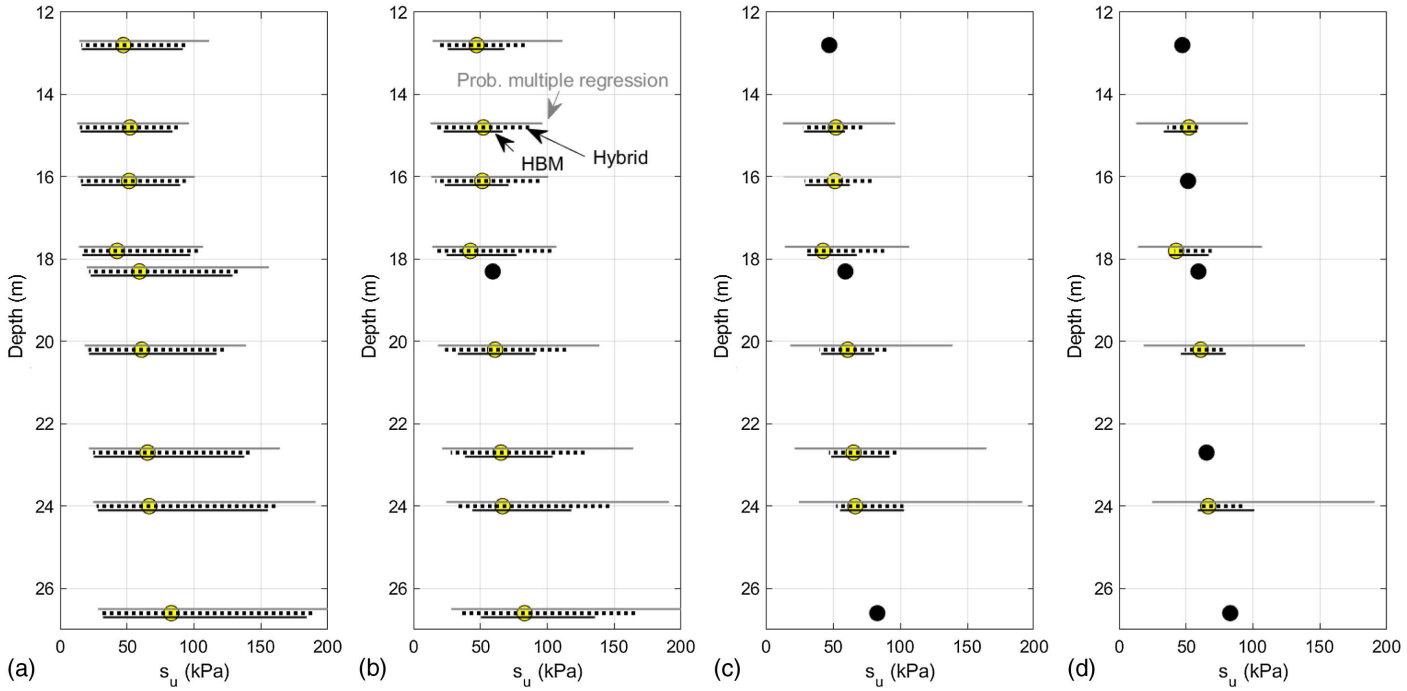


Fig. 17. Multivariate inference results for (a) Case 1; (b) Case 2; (c) Case 3; and (d) Case 4.

Table 3. Computer runtimes (in minutes) for the real examples

Method	Multivariate cases ($n = 7$)	Bivariate cases ($n = 2$)	Multivariate cases ($n = 7$ with reduced $\mathbf{X}^o$)
HBM	90.2	22.7	47.4
Hybridization	66.9	17.6	39.2

key constraints to keep this research practically meaningful. The constraints are working with actual soil/rock databases and with algorithms that can address these databases at affordable computational cost.

Data Availability Statement

All database data, case history data, and computer codes that support the findings of this study are available from the corresponding author upon reasonable request.

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References

Behmanesh, I., B. Moaveni, G. Lombaert, and C. Papadimitriou. 2015. "Hierarchical Bayesian model updating for structural identification." *Mech. Syst. Sig. Process.* 64–65 (Dec): 360–376. https://doi.org/10.1016/j.ymssp.2015.03.026.

Bozorgzadeh, N., and R. J. Bathurst. 2019. "Hierarchical Bayesian approaches to statistical modelling of geotechnical data." *Georisk* 1–18. https://doi.org/10.1080/17499518.2020.1864411.

Bozorgzadeh, N., J. P. Harrison, and M. D. Escobar. 2019. "Hierarchical Bayesian modelling of geotechnical data: Application to rock strength." *Géotechnique* 69 (12): 1056–1070. https://doi.org/10.1680/jgeot.17.P.282.

Ching, J. 2020. "Value of geotechnical big data and its application in site-specific soil property estimation." *J. GeoEng.* 15 (4): 173–182. https://dx.doi.org/10.6310%2fjog.202012_15(4).1.

Ching, J. 2018. "What does the soil parameter estimated from a transformation model really mean?" *J. GeoEng.* 13 (3): 105–113. https://dx.doi.org/10.6310%2fjog.201809_13(3).2.

Ching, J., and K. K. Phoon. 2014a. "Correlations among some clay parameters—The multivariate distribution." *Can. Geotech. J.* 51 (6): 686–704. https://doi.org/10.1139/cgj-2013-0353.

Ching, J., and K. K. Phoon. 2014b. "Transformations and correlations among some parameters of clays—The global database." *Can. Geotech. J.* 51 (6): 663–685. https://doi.org/10.1139/cgj-2013-0262.

Ching, J., and K. K. Phoon. 2019. "Constructing site-specific probabilistic transformation model by Bayesian machine learning." *J. Eng. Mech.* 145 (1): 04018126. https://doi.org/10.1061/(ASCE)EM.1943-7889.0001537.

Ching, J., and K. K. Phoon. 2020. "Constructing a site-specific multivariate probability distribution using sparse, incomplete, and spatially variable (MUSIC-X) data." *J. Eng. Mech.* 146 (7): 04020061. https://doi.org/10.1061/(ASCE)EM.1943-7889.0001779.

Ching, J., K. K. Phoon, Y. H. Ho, and M. C. Weng. 2020a. "Quasi-site-specific prediction for deformation modulus of rock mass." *Can. Geotech. J.* https://doi.org/10.1139/cgj-2020-0168.

Ching, J., K. K. Phoon, Z. Khan, D. M. Zhang, and H. W. Huang. 2020b. "Role of municipal database in constructing site-specific multivariate probability distribution." *Comput. Geotech.* 124 (Aug): 103623. https://doi.org/10.1016/j.compgeo.2020.103623.

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Ching, J., K. K. Phoon, Z. Yang, and A. W. Stuedlein. 2021. "Constructing a quasi-site-specific multivariate probability distribution model for soil properties using sparse, incomplete, and three-dimensional (MUSIC-3X) data." *Georisk*. https://doi.org/10.1080/17499518.2021.1896005.

Chung, Y., A. Gelman, S. Rabe-Hesketh, J. Liu, and V. Dorie. 2015. "Weakly informative prior for point estimation of covariance matrices in hierarchical models." *J. Educ. Behav. Stat.* 40 (2): 136–157. https://doi.org/10.3102/1076998615570945.

Gelman, A., and J. Hill. 2006. *Data analysis using regression and multilevel/hierarchical models*. New York: Cambridge University Press.

Geman, S., and D. Geman. 1984. "Stochastic relaxation, Gibbs distribution and the Bayesian restoration of images." *IEEE Trans. Pattern Anal. Mach. Intell.* PAMI-6 (6): 721–741. https://doi.org/10.1109/TPAMI.1984.4767596.

Gilks, W. R., D. J. Spiegelhalter, and S. Richardson. 1996. *Markov chain Monte Carlo in practice*. London: Chapman and Hill.

Huang, A., and M. P. Wand. 2013. "Simple marginally noninformative prior distributions for covariance matrices." *Bayesian Anal.* 8 (2): 439–452. https://doi.org/10.1214/13-BA815.

James, A. 1964. "Distributions of matrix variates and latent roots derived from normal samples." *Ann. Math. Stat.* 35 (2): 475–501. https://doi.org/10.1214/aoms/1177703550.

Johnson, N. L. 1949. "Systems of frequency curves generated by methods of translation." *Biometrika* 36 (1–2): 149–176. https://doi.org/10.2307/2332539.

Kulhawy, F. H., and P. W. Mayne. 1990. *Manual on estimating soil properties for foundation design*. Rep. No. EL6800. Palo Alto, CA: Electric Power Research Institute.

Lu, S., J. Zhang, S. Zhou, and A. Xu. 2018. "Reliability prediction of the axial ultimate bearing capacity of piles: A hierarchical Bayesian method." *Adv. Mech. Eng.* 10 (11): 168781401881105. https://doi.org/10.1177/1687814018811054.

Mardia, K. V., J. T. Kent, and J. M. Bibby. 1979. *Multivariate analysis*. London: Academic Press.

Mayne, P. W., B. R. Christopher, and J. DeJong. 2001. *Manual on subsurface investigations*. National Highway Institute Publication No. FHWA NHI-01-031. Washington, DC: Federal Highway Administration.

Nagel, J. B., and B. Sudret. 2016. "A unified framework for multilevel uncertainty quantification in Bayesian inverse problems." *Probab. Eng. Mech.* 43 (Jan): 68–84. https://doi.org/10.1016/j.probengmech.2015.09.007.

Ou, C. Y., and J. T. Liao. 1987. *Geotechnical engineering research report*. Rep. No. GT96008. Taipei, Taiwan: National Taiwan Univ. of Science and Technology.

Phoon, K. K. 2020. "The Goldilocks dilemma—Too little or too much data." *GeoStrata* 24 (1): 14–16.

Phoon, K. K., and J. Ching. 2021. "Project deepGeo—Data-driven 3D subsurface mapping." *J. GeoEng.* 16 (2): 61–73.

Phoon, K. K., J. Ching, and T. Shuku. 2021. "Challenges in data-driven site characterization." In *Georisk*. Milton Park, MI: Taylor & Francis. https://doi.org/10.1080/17499518.2021.1896005.

Phoon, K. K., J. Ching, and Y. Wang. 2019. "Managing risk in geotechnical engineering—From data to digitalization." In *Proc., 7th Int. Symp. on Geotechnical Safety and Risk (ISGSR 2019)*. London: International Society for Soil Mechanics and Geotechnical Engineering.

Wu, S., P. Angelikopoulos, J. L. Beck, and P. Koumoutsakos. 2019. "Hierarchical stochastic model in Bayesian inference for engineering applications: Theoretical implications and efficient approximation." *ASCE-ASME J. Risk Uncertainty Eng. Syst. Part B Mech. Eng.* 5 (1): 011006. https://doi.org/10.1115/1.4040571.

Yu, J.-Z., and H. Baroud. 2019. "Quantifying community resilience using hierarchical Bayesian kernel methods: A case study on recovery from power outages." *Risk Anal.* 39 (9): 1930–1948. https://doi.org/10.1111/risa.13343.

Zhang, J., C. H. Juang, J. R. Martin, and H. W. Huang. 2016. "Inter-region variability of Robertson and Wride method for liquefaction hazard analysis." *Eng. Geol.* 203 (Mar): 191–203. https://doi.org/10.1016/j.enggeo.2015.12.024.

Zhang, J., J. P. Li, L. M. Zhang, and H. W. Huang. 2014. "Calibrating cross-site variability for reliability-based design of pile foundations." *Comput. Geotech.* 62 (Oct): 154–163. https://doi.org/10.1016/j.compgeo.2014.07.013.

Zhang, L. 2016. *Engineering properties of rocks*. 2nd ed. Cambridge, MA: Elsevier.

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